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跨域港口與機場目標偵測：YOLO11 模組變體、資料增強與圖形化介面研究支援系統

Cross-Domain Harbor–Airport Object Detection: YOLO11 Module Variants, Data Augmentation, and GUI-Based Research Support Systems

研 究 生： 洪紹鈞 (Hung, Shao-Chun)
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跨域港口與機場目標偵測：YOLO11 模組變體、資料增強與圖形化介面研究支援系統

學生：洪紹鈞　　指導教授：吳金典 博士

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摘　要

港口與機場影像在目標尺度、密度、背景及語意共現關係上具有明顯差異。本研究建立一套整合四類港口目標與三類機場目標的七類跨域偵測設定，以 YOLO11n 為基準，比較 EfficientViM Block、DySample 與自適應空間特徵融合（ASFF）三種架構介入方式。比較結果顯示，各架構在不同任務、類別與指標之間各有取捨，沒有單一架構普遍占優。後續錯誤分析指出，船舶標註不完整、港口車輛監督不足，以及船載航空器情境缺失，皆是僅調整模型架構無法處理的資料問題。

在固定評估基準下，修正部分船舶訓練標註後，YOLO11n 的 mAP@50 由 76.2% 提升至 77.0%，其他船舶的 AP@50 由 41.9% 增至 45.0%。其後的七類實驗以修正後的資料設定為基準，並在單一港口領域監督處理中，同時加入港口車輛人工標註、港口車輛前景合成及船載航空器前景合成。與未加入上述資料的基準相比，三種架構在完整測試集上的 mAP@50、mAP@50–95，以及大型與小型車輛 AP@50 均有所提升。在港口特定評估中，三種架構的大型車輛偵測皆由近乎無法偵測提升為非零且可量測的表現；小型車輛與標準測試集中飛機的結果則仍因架構而異。由於三類監督同時加入且未進行組件式消融，本研究無法分離其個別因果效果；此外，最終七類實驗的模型檢查點未能用於建構式船載航空器測試，因此該測試尚無對應的最終結果。

本研究亦建置兩套圖形化介面，支援資料建置、標註審查、模型推論、視覺化及報告產生。獨立確認結果顯示，當研究人員提供船舶參考框後，供框式船種輔助器可產生具實用性的候選類別，協助審查全圖偵測流程中遺漏或存疑的標註。此成果支持將該工具定位為保留人工最終決策權的漏標補標輔助，而非自動標註器或偵測器的替代方案。整體而言，跨域目標偵測表現除受模型架構影響外，也取決於標註品質、任務相關監督是否充分，以及訓練資料能否呈現原本缺失的場景與目標共現關係。

關鍵字：YOLO11、跨域目標偵測、資料中心方法、標註修正、前景合成、圖形化介面研究支援

Cross-Domain Harbor–Airport Object Detection: YOLO11 Module Variants, Data Augmentation, and GUI-Based Research Support Systems

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Abstract

Harbor and airport imagery differs in scale, density, background, and semantic co-occurrence. A seven-class setting integrates four harbor and three airport-derived classes. YOLO11n, EfficientViM Block, DySample, and Adaptive Spatial Feature Fusion (ASFF) showed task-, class-, and metric-dependent trade-offs with no universally best architecture. Error analysis linked residual failures to incomplete vessel annotations, inadequate harbor-vehicle supervision, and missing shipboard-aircraft contexts.

With fixed evaluation labels, partial vessel-label correction raised YOLO11n mAP@50 from 76.2% to 77.0% and other-vessel AP@50 from 41.9% to 45.0%. A combined harbor-domain treatment then added manual and foreground-composited harbor vehicles together with foreground-composited shipboard aircraft. All architectures improved mAP@50, mAP@50–95, and both vehicle-class AP@50 values on the full test set. On a small harbor-specific evaluation, large-vehicle detection became measurable for all three architectures, whereas small-vehicle and airplane results varied. Because the components were introduced together, their individual effects cannot be isolated; final checkpoints were unavailable for the constructed shipboard evaluation.

Two graphical interfaces support data construction, review, inference, visualization, and reporting. Independent confirmation showed that, once a researcher supplies a vessel reference box, the supplied-box vessel-type assistant can provide useful class candidates for reviewing annotations missed or left uncertain by the full-image detection workflow. This result supports its role as a human-confirmed annotation aid rather than an autonomous label writer or a replacement for the detector. Overall, cross-domain detection depended on architecture, annotation quality, task-relevant supervision, and representation of missing contexts.

Keywords: YOLO11; cross-domain object detection; data-centric detection; annotation correction; foreground compositing; GUI-based research support

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Chapter 1. Introduction

1.1 Research Background and Problem Setting

Image-based object detection provides a common means of locating and classifying targets in harbor and airport remote-sensing imagery. Harbor scenes contain substantial variation in vessel scale, orientation, density, and visual background, particularly when small ships occur near ports or in closely spaced groups [1]. Airport imagery presents related scale and density challenges, but its objects occur among runways, aprons, terminal structures, and service areas rather than water, vessels, and docks [2]. The task considered in this thesis therefore combines the harbor classes `naval`, `merchant`, `dock`, and `other_vessel` with the airport-derived classes `airplane`, `large-vehicle`, and `small-vehicle`. A detector trained in this integrated seven-class setting must learn from source populations that differ in scene context, class distribution, object scale, background, and semantic co-occurrence.

These domain differences are accompanied by three distinct data-side limitations. The first is annotation incompleteness or inconsistency. Objects may be visible in harbor imagery while their boxes or class labels are missing or inconsistent, so the available annotations may not fully represent the visual content used for training and evaluation. Missing instance annotations can introduce unreliable supervision in fully supervised object detection [3, 4]. The annotation audit conducted in this study identified this problem for vessel records and made an audited, fixed evaluation reference necessary before the effect of training-label correction could be examined.

The second limitation is insufficient class and domain supervision. Vehicles of interest are already visible in harbor images, but the source labels provide limited supervision for the two vehicle classes. The third is an absent target-context relationship: the harbor source contains no `airplane` class and therefore no co-occurrence of aircraft with vessels or decks. Airport imagery supplies aircraft and vehicle examples, but pavement-based examples do not reproduce a shipboard context. Because unsuitable surrounding context can reduce the value of inserted objects [5], this relationship cannot be treated as label-space integration alone. Prior shipboard-aircraft and synthetic-aircraft research establishes the relevance of the broader

maritime scene while also motivating a clear distinction between constructed and independently acquired imagery [6, 7]. The research problem is consequently both cross-domain and supervision-dependent: it requires data-quality correction for existing annotations, additional supervision for underrepresented harbor-domain objects, and construction of a target-context relationship absent from the source data.

1.2 Research Motivation

The research began with an architecture-focused investigation and subsequently shifted toward data quality and supervision. EfficientViM Block, DySample, and Adaptive Spatial Feature Fusion (ASFF) were introduced separately into YOLO11n at the stages of high-level feature representation, top-down upsampling, and multi-scale fusion. Across the evaluated settings, the resulting comparisons showed trade-offs that depended on the task, class, metric, and computational cost. No architecture provided a consistent advantage across all evaluated tasks and metrics, so the comparison did not justify selecting one modification as a universal solution.

Because no architecture was consistently superior, the study turned to detector-error inspection rather than extending the architecture search. This inspection exposed limitations in annotation quality and supervision that model modification alone could not address. Some visible vessels required human-confirmed annotation correction, vehicles already present in harbor scenes needed stronger class supervision, and the intended shipboard-aircraft co-occurrence was absent from the harbor source. These observations led to two formal data-centric comparisons: partial correction of vessel training annotations under Conditions A and B, followed by a combined V0/V1 treatment that added harbor-vehicle supervision and constructed shipboard-aircraft examples. GUI-based systems supported dataset construction, annotation review and assistance, inference, visualization, analysis, and reporting throughout this workflow.

1.3 Method and Evaluation Overview

The study follows three connected methodological lines. First, the model-side comparison uses YOLO11n as the baseline and evaluates EfficientViM Block, DySample, and ASFF individually and in two- and three-module combinations [8]. EfficientViM Block modifies high-level feature representation, DySample replaces fixed top-down upsampling with content-dependent sampling, and ASFF modifies prediction-stage multi-scale fusion. The comparison

characterizes task-dependent behavior rather than seeking one architecture that dominates every class and metric.

Second, the annotation-quality line audits the original vessel annotations and establishes a corrected, frozen evaluation reference. The historical pre-audit record is used to motivate this control, not as a separate causal experiment. With the evaluation annotations held fixed, detectors trained on the original vessel labels are compared with detectors trained on the partially corrected, human-confirmed vessel labels. This Condition A/Condition B comparison isolates the change in training-label state from changes to the evaluation reference and examines both aggregate performance and detector-error behavior.

Third, the targeted-supervision line begins from a corrected seven-class harbor–airport setting that combines the corrected harbor vessel-label state with the airport-derived classes. V0 is the corrected seven-class baseline: it contains no added harbor vehicle supervision and no shipboard-aircraft foreground-composited training samples. V1 starts from the same setting and simultaneously adds manual harbor-vehicle annotations, foreground-composited harbor vehicles, and foreground-composited shipboard aircraft with synchronized bounding-box annotations, following the established compositing principle [9]. The single V0/V1 comparison evaluates this combined targeted harbor-domain supervision package globally and examines its vehicle and airplane responses in the relevant evaluation settings. Because the completed design contains no component-wise ablation, the three additions cannot be assigned separate causal effects.

Two GUI-based research-support systems connect these methods to the working process. The dataset system supports data inspection, annotation review, foreground compositing, synchronized label generation, and a vessel-type assistant that presents ranked candidates for human confirmation. The post-training platform supports registered-model inference, prediction visualization, result inspection, auxiliary analysis, and report export. These systems support the construction, review, and analysis stages surrounding the detector experiments.

1.4 Research Questions and Objectives

The thesis addresses the following four research questions:

1. **RQ1 – Architecture behavior:** What task-dependent performance trade-offs are observed among YOLO11n and the EfficientViM Block, DySample, and ASFF variants

across the evaluated detection settings?

2. **RQ2 – Annotation quality and correction:** How does partial correction of vessel training annotations affect detector performance and error behavior when the evaluation annotations are fixed?

3. **RQ3 – Combined harbor-domain supervision:** How does a combined targeted harbor-domain supervision treatment affect the corrected seven-class detector globally and in the harbor-specific vehicle evaluation, and what can be concluded about the constructed shipboard-aircraft setting?

4. **RQ4 – Research-support systems:** To what extent do the implemented GUI-based research systems support data construction, annotation review, detection analysis, and human-confirmed vessel-type assistance within the evaluated workflow?

RQ3 concerns one combined V1 treatment containing three types of added supervision. The corresponding research objectives are:

1. compare the YOLO11n baseline with the three retained architecture interventions and characterize their task-, class-, and metric-dependent behavior;

2. audit vessel annotations and evaluate partial training-label correction under a shared corrected and frozen evaluation reference;

3. evaluate the combined harbor-domain supervision package on the global and harbor-specific tests; document what can be evaluated in the constructed shipboard-aircraft setting without assigning component-level effects; and

4. implement and verify GUI-based support for data construction, annotation assistance, inference, visualization, analysis, and reporting.

1.5 Contributions and Thesis Organization

The thesis makes four high-level contributions:

1. **Cross-domain seven-class data setting.** It establishes a unified harbor–airport detection setting that preserves the distinction between four harbor classes and three airport-derived

classes. Deterministic source-data recounts and documented evaluation records make the scope and provenance of this setting explicit.

2. **Data-centric correction and construction workflow.** It distinguishes annotation quality, insufficient supervision for vehicles already visible in harbor imagery, and a missing shipboard-aircraft target-context representation, and addresses them through human-confirmed correction or synchronized data construction.

3. **Comparative experiments.** It reports task-dependent architecture trade-offs, the effects of partial vessel training-label correction under a fixed reference, and global, harbor-vehicle, and airplane responses under the combined V0/V1 targeted-supervision treatment.

4. **GUI-based research support.** It implements two complementary systems for dataset construction, annotation review, detector inference and visualization, auxiliary analysis, and reporting, including a vessel-type assistant that supplies ranked candidates subject to human confirmation.

The remainder of the thesis follows a six-chapter structure. Chapter 2, *Literature Review*, reviews cross-domain detection, the selected architecture interventions, annotation quality, foreground compositing, contextual placement, and related shipboard-aircraft and synthetic-data research. Chapter 3, *Methodology*, defines the YOLO11n baseline, the EfficientViM Block, DySample, and ASFF variants, together with the foreground-compositing and annotation-generation method. Chapter 4, *Software Tools for Data Preparation and Post-Training Detection Analysis*, presents the dataset and annotation tools, the vessel-type assistant, and the post-training inference and result-inspection platform. Chapter 5, *Experiments and Results*, reports the architecture comparison, annotation-correction experiment, combined V0/V1 harbor-domain supervision experiment, and evaluation of the supporting systems. Chapter 6, *Conclusion and Future Work*, synthesizes the findings, limitations, and directions for future research.

Chapter 2. Literature Review

2.1 Harbor–Airport Detection Across Domains

The original YOLO formulation established one-stage detection as the direct prediction of object classes and bounding boxes from a complete image in a single network evaluation [10]. This approach motivates the detector family used throughout the thesis. The experiments use the nano-scale YOLO11 detector as their shared baseline [8]; Chapters 3 and 5 specify its architecture, version, initialization, and training configuration.

Optical remote-sensing images of harbors combine substantial variation in vessel scale, orientation, aspect ratio, and density with backgrounds that range from open water to congested ports. ShipRSImageNet documents these conditions across multiple sensors, ports, resolutions, and environments, and its benchmarks identify small, densely grouped ships as a continuing difficulty [1]. Its hierarchy also separates warships, merchant ships, other vessels, and docks. The dock category is relevant because docks are fixed infrastructure rather than vessels, yet their rectangular appearance can interfere with near-shore ship detection [1]. Harbor detection therefore requires both fine-grained discrimination among visually related vessels and separation of vessels from structurally similar infrastructure.

Airport and aerial imagery share the problems of large scale variation, dense object arrangements, and arbitrary orientation. DOTA represents these conditions with oriented annotations across categories that include planes, large vehicles, and small vehicles [2]. The airport component used in this research draws on those categories and on GDIT aircraft imagery, following the preceding aerial-data study in this research line [11, 12]. Its scene context is nevertheless different: aircraft and service vehicles are associated with runways, aprons, markings, and terminal areas rather than water, docks, and vessels.

The shared visual challenges do not make the two source populations interchangeable. The harbor taxonomy distinguishes vessels by operational type and separates them from fixed dock infrastructure, whereas the airport taxonomy distinguishes aircraft from two vehicle-size groups. These classes also occupy different scene roles: service vehicles may form dense

groups around aircraft, while vessels occur over water or beside port structures. A joint detector must therefore learn one label space from source populations that differ both statistically and semantically, making domain-group and class-specific analysis more informative than an undifferentiated description of the combined data.

Combining the harbor and airport label spaces creates a cross-domain detection setting rather than a homogeneous dataset. The two sources differ in background, object distribution, scale relationships, support surfaces, and semantic co-occurrence. Domain-adaptive detection addresses related discrepancies through image- and instance-level alignment [13]; this study instead examines direct data integration and the explicit construction of a missing target context. Airport data supplies examples of aircraft, but the original harbor source contains no aircraft located on or operating from a vessel. Label integration alone therefore cannot represent the intended shipboard-aircraft relationship.

2.2 Feature Enhancement and Multi-Scale Fusion

Feature pyramids provide the conceptual background for the three architecture interventions examined in this thesis. A convolutional hierarchy reduces spatial resolution while increasing semantic abstraction, whereas Feature Pyramid Networks use top-down and lateral connections to construct semantically stronger features at several resolutions [14]. The resulting levels have complementary characteristics: high-resolution features retain finer localization information, while lower-resolution features represent broader context.

EfficientViM is a lightweight vision architecture based on Hidden State Mixer State Space Duality. Its central idea is to move costly channel-mixing projections and gating into a compressed hidden-state representation while combining global interaction with local processing [15]. This feature-mixing principle is relevant to the high-level representation used by a detector. The source architecture motivates the feature-representation intervention examined in this thesis; the implemented block and its differences from the complete EfficientViM architecture are specified in Chapter 3.

Top-down feature paths must also increase the spatial resolution of low-resolution semantic features before fusion. Fixed nearest-neighbor or bilinear interpolation uses predetermined sampling geometry regardless of image content. DySample instead predicts content-dependent offsets and performs point resampling through standard grid sampling, avoiding the dynamic

convolution and custom operators used by heavier kernel-based upsamplers [16]. This mechanism is relevant where upsampling must retain spatial information for targets represented at different scales.

ASFF addresses a later stage of the pyramid. Because corresponding locations at different levels can provide conflicting or complementary information, ASFF aligns the levels to a target resolution and learns spatially varying weights for their contributions [17]. It therefore offers a data-driven way to combine localization detail and higher-level semantics before prediction.

These three methods address complementary stages rather than competing versions of one solution. EfficientViM motivates a change in feature representation, DySample changes the geometry of upsampling, and ASFF changes multi-scale fusion. Their source mechanisms make each a reasonable intervention for a detector operating across varied object scales and contexts, while their effects in the implemented YOLO11n variants remain questions for the experiments.

2.3 Synthetic Data Augmentation

Foreground compositing is an established mechanism for increasing instance diversity or density. Ghiasi et al. found that randomly pasting segmented instances can improve instance-segmentation and bounding-box performance without an explicit context model in the COCO and LVIS settings [9]. Dwibedi et al. constructed training images by placing cut-out objects on unrelated backgrounds and showed that boundary artifacts and blending choices materially affect the resulting detection signal [18]. In maritime SAR imagery, BoxPaste crops ships using bounding-box annotations and inserts them into other training images to increase object density [19]. Together, these studies support compositing as a data-augmentation principle while showing that source–host conditions, boundaries, and annotations influence its usefulness.

These precedents also distinguish two data-construction purposes. Within-dataset compositing can increase the frequency or diversity of object instances already represented by a benchmark, as in random Copy-Paste and BoxPaste [9, 19]. Cross-domain compositing can instead introduce a semantic co-occurrence missing from the source data. For the latter purpose, the central issue is not whether pasting works in general, but whether the foreground, host, and placement jointly provide a coherent training example of the absent relationship.

Placement context adds a separate consideration. Dvornik et al. showed that inserting objects into unsuitable surroundings can reduce detection performance and that a learned context

model can select more appropriate regions when labeled data are scarce [5]. This finding is compatible with random Copy-Paste because the studies use different datasets, source–target relationships, and augmentation regimes [9]. Together, they indicate that the importance of placement strategy depends on how strongly the inserted object and host scene differ. The airport-to-harbor transfer in this thesis changes the support surface, surrounding objects, and plausible scale relationships, which motivates constraining placement in the present setting. Chapter 3 defines the operator and software roles of that constraint.

The workflow also follows a direct methodological lineage. The preceding aerial-data study used Spectral Matting to obtain object foregrounds and a graphical interface to blend them into airport imagery while generating corresponding labels [20, 12]. The present study transfers this workflow from aerial or airport hosts to optical harbor scenes and uses operator-guided placement to construct the missing shipboard relationship. The constructed-domain evaluation is considered separately from the standard test because the final experiment checkpoints were not available for that evaluation.

Shipboard aircraft and synthetic aviation data both have relevant precedents. Zhu et al. studied carrier-based-aircraft detection on simulated three-dimensional carrier scenes with a deep neural network [6], while Fan et al. addressed multi-scale object detection on an aircraft-carrier surface with an improved Faster R-CNN [21]. In an adjacent task, Guo et al. generated virtual flight-deck operations because real trajectory data were scarce, although their objective was operation recognition rather than image-based bounding-box detection [22]. RarePlanes combines real satellite imagery with a large synthetically generated aircraft corpus and evaluates the contribution of synthetic data to overhead aircraft detection and classification [7]. These studies establish both the relevance of the shipboard-aircraft scene and the use of synthetic data when aviation imagery is scarce.

Synthetic data construction does not remove the need to audit the original annotations. Niitani et al. describe sparsely annotated detection data as a reliability problem because missing object boxes can create false training signals, while Xu et al. examine how missing instance-level annotations affect fully supervised object detection [3, 4]. These findings motivate human-reviewed inspection and correction when visible harbor objects lack required boxes. In this thesis, annotation correction is a supporting data-quality process distinct from foreground compositing: the latter introduces a missing cross-domain context, whereas the former repairs incomplete supervision for objects already present. Chapter 5 evaluates partial vessel training-

label correction under a fixed corrected evaluation reference and reports architecture-dependent detector responses.

The evaluation domain determines how the result can be interpreted. Measurable detection performance on constructed shipboard imagery would show a response within that constructed context, but it would not by itself demonstrate generalization to independently acquired real shipboard imagery. The later experimental design therefore treats constructed and real imagery as separate evaluation domains.

2.4 Literature Synthesis and Research Positioning

Together, the literature defines the problem addressed in this thesis. The harbor source lacks the target co-occurrence of aircraft and vessels, while airport aircraft examples do not reproduce harbor backgrounds, vessel support surfaces, or the scale relationships of a shipboard scene. Missing or inconsistent annotations can also provide incomplete supervision, which motivates human-reviewed data-quality control. Foreground compositing offers an established way to introduce missing instances, but the large difference between source objects and host scenes makes placement control important. Results from the constructed context must still be interpreted separately from real-domain transfer.

Table 1: How representative prior work informs the present study.

Study	Relevant finding or approach	Relevance to the present study	Difference from the present study
Lee et al.; Liu et al. [15, 16, 17]	Efficient representation, learned point upsampling, and spatial fusion target distinct processing stages.	Motivate three complementary architecture interventions.	Each source idea is adapted separately to a common YOLO11n baseline.
Niitani et al.; Xu et al. [3, 4]	Sparse or missing instance annotations can impair fully supervised detector training.	Motivate human-reviewed annotation audit and correction.	Chapter 5 evaluates partial training-label correction under a fixed corrected evaluation reference.
Ghiasi et al. [9]	Random instance compositing can improve segmentation and box detection without a context model.	Establishes foreground compositing as a valid augmentation principle.	Aircraft are transferred across source domains with user-guided placement.
Dvornik et al. [5]	Context suitability can determine whether object insertion helps or harms detection.	Motivates explicit control of insertion location.	The allowable region is operator-defined rather than learned.
Suo et al. [19]	SAR ship crops are pasted into other training images to increase object density.	Provides maritime precedent for paste-based augmentation.	The transfer uses aircraft foregrounds and optical harbor hosts.
Zhu et al. [6]	A deep detector is studied on simulated three-dimensional carrier scenes.	Establishes shipboard-aircraft detection precedent.	The present scenes are two-dimensional cross-dataset composites in a joint detector.
Fan et al. [21]	Multi-scale carrier-surface objects are detected with an improved Faster R-CNN.	Establishes carrier-surface scale and scene relevance.	The present focus is cross-domain data construction within seven classes.
Shermeyer et al. [7]	Real and synthetic satellite aircraft data are evaluated together and separately.	Supports synthetic aviation data and explicit evaluation boundaries.	The constructed shipboard domain is isolated from the standard test.
Cai [12]	A GUI supports aerial-object extraction, compositing, and label generation.	Provides the preceding data and workflow lineage.	The target changes to harbor hosts with context-constrained placement.

Architecture, context construction, and annotation correction therefore play different roles. EfficientViM, DySample, and ASFF motivate complementary changes to representation, upsampling, and fusion, but feature processing cannot supply a semantic context absent from the training imagery. Foreground compositing constructs that missing context directly. Annotation review instead addresses missing or inconsistent boxes for visible objects; Chapter 5 evaluates the resulting partial training-label correction against the same fixed corrected evaluation reference. The intended evaluation design distinguishes standard-domain evidence from a separate constructed-domain test, because the two domains support different claims.

Accordingly, the thesis studies joint seven-class behavior, three architecture interventions, human-confirmed partial annotation correction, and construction of the missing shipboard-aircraft context. The standard-domain results are reported separately from the constructed-domain test, for which no result matched to the final checkpoints is available. Chapters 4 and 5 evaluate the supporting software independently of the detector comparisons. This problem-oriented scope follows directly from the identified data-quality and missing-context limitations.

Chapter 3. Methodology

3.1 Model and Data Methods Overview

This chapter defines the model-side and data-side methods used in the study. The model-side method uses YOLO11n as a common baseline and introduces three interventions: EfficientViM Block for high-level feature representation, DySample for top-down sampling, and Adaptive Spatial Feature Fusion (ASFF) for multi-scale fusion. Chapter 5 evaluates these interventions individually and in combination. The data-side method is a class-agnostic foreground-compositing and annotation-generation pipeline used to add harbor-vehicle supervision and construct the missing shipboard-aircraft context. Chapter 5 reports the dataset versions, experimental controls, and results, while Chapter 4 describes the interface through which the data-side method is applied.

3.2 YOLO11n Baseline

YOLO11 is an Ultralytics one-stage object-detection family with multiple model scales and task variants [8]. This study uses the nano-scale detector, YOLO11n, as the common baseline for all standalone architecture variants. Each variant starts from the same publicly released YOLO11n pretrained initialization so that the inserted module, rather than a different initialization scheme, defines the architectural change.

The baseline backbone uses convolutional downsampling and C3k2 feature blocks, followed by a Spatial Pyramid Pooling-Fast (SPPF) module and a C2PSA attention block. Its neck combines top-down upsampling, lateral concatenation, and bottom-up aggregation. The detection head receives three feature levels: P3/8, the highest-resolution level; P4/16, the intermediate level; and P5/32, the lowest-resolution and most semantically abstract level. These three tensors provide the insertion context for the modifications defined in Sections 3.3.1–3.3.3. Figure 1 summarizes the baseline data flow.

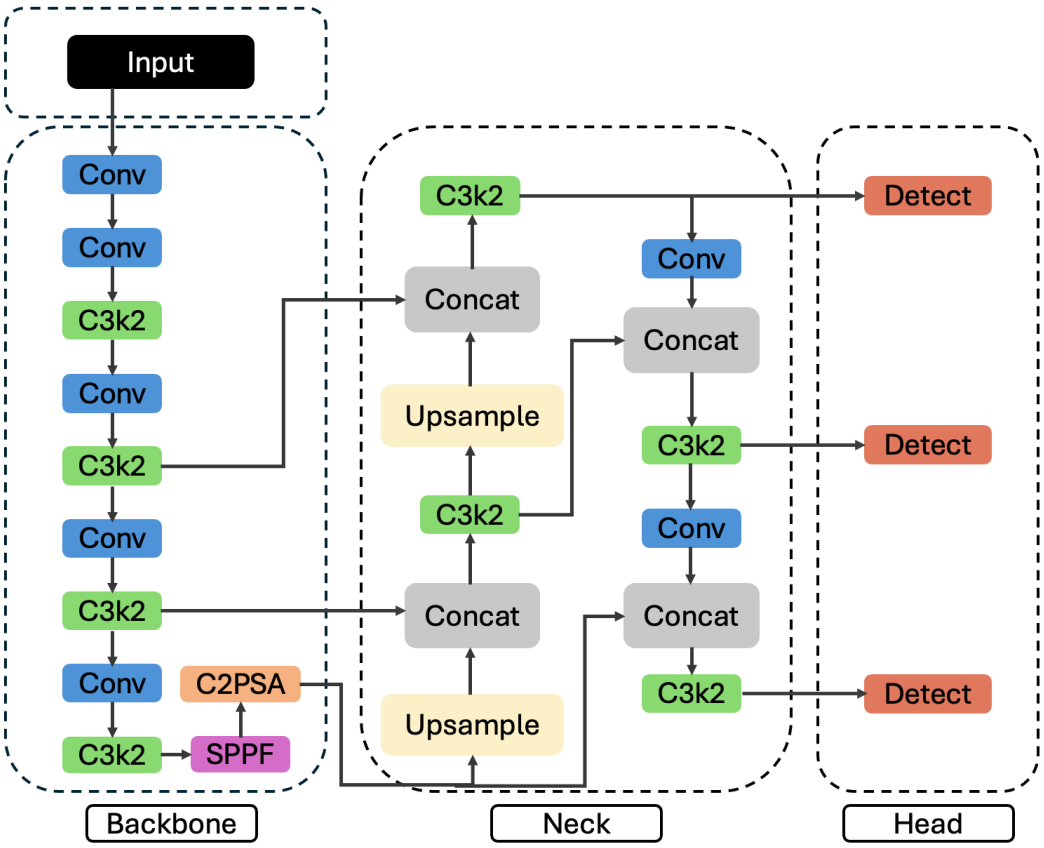


Figure 1: Author-generated schematic of the implemented YOLO11n baseline and its P3/8, P4/16, and P5/32 detection features.

3.3 YOLO11n Architecture Interventions

The following subsections define the three architecture interventions individually. Each changes one functional stage while leaving the remainder of the YOLO11n pipeline unchanged. The EfficientViM block modifies high-level feature representation, DySample modifies top-down sampling, and ASFF changes multi-scale fusion before prediction. The experiments include both standalone and combined configurations. Standalone configurations isolate each insertion concept, while two- and three-module combinations test whether their effects are additive.

3.3.1 EfficientViM Feature-Mixing Block (P5-Only)

EfficientViM is a lightweight vision architecture based on Hidden State Mixer State Space Duality (HSM-SSD) [15]. Its source design performs channel-mixing projections and gating in a compressed hidden-state representation while combining global interaction with local depth-

wise processing. The implementation in this study adapts that feature-mixing idea as a block inside a YOLO11 C3k-style module; it is not a replacement of the complete YOLO11 backbone by the original EfficientViM architecture.

The adapted unit contains four sub-operators, each wrapped by a gated residual. Let $x_i \in \mathbb{R}^{C\times H'\times W'}$ be the input to sub-operator F_i, let $\alpha_i \in \mathbb{R}^C$ be a learnable channel parameter, let $\odot$ denote channel-wise multiplication, and let the logistic sigmoid be $\sigma(\alpha) = \frac{1}{1+e^{-\alpha}}$. The update is

$$x_{i+1} = \big(1 - \sigma(\alpha_i)\big) \odot x_i + \sigma(\alpha_i) \odot F_i(x_i). \tag{3.1}$$

Because $\sigma(\alpha_i) \in (0, 1)$, Equation (3.1) performs a channel-wise convex interpolation between the identity path and the transformed feature. As $\sigma(\alpha_i)$ approaches zero the identity path dominates, whereas values approaching one emphasize $F_i(x_i)$. Each α_i is initialized to 10^{-4}, giving $\sigma(\alpha_i) \approx 0.500025$ at initialization. Although the implementation comments label these parameters as LayerScale, the executed operation is the sigmoid interpolation in Equation (3.1), not an additive $x + \alpha F(x)$ residual.

The four transformations preserve the input shape. F_1 and F_3 are local 3×3 depthwise convolutions; F_2 is the flatten–normalize–HSM-SSD path for global hidden-state mixing; and F_4 is a feed-forward network with expansion ratio four. In the implemented wrapper, a 384-channel input is projected to 256 channels and split into two 128-channel paths. Within the processed C3k path, a 64-channel half passes through two stacked EfficientViM units. The retained and processed paths are concatenated and projected to a 256-channel output. The mixing step uses a 64-dimensional hidden state, and the feed-forward path expands from 64 to 256 channels before projecting back to 64.

Although the wrapper exposes several neck positions, the EfficientViM-containing branch is active only in the final P5/32 neck-output block; earlier positions retain ordinary bottlenecks. The standalone variant is therefore P5-only and keeps the conventional detection head. No placement ablation was performed, so P5 is not claimed to be optimal.

Figure 2 summarizes the adapted unit, its channel flow, and its effective insertion point.

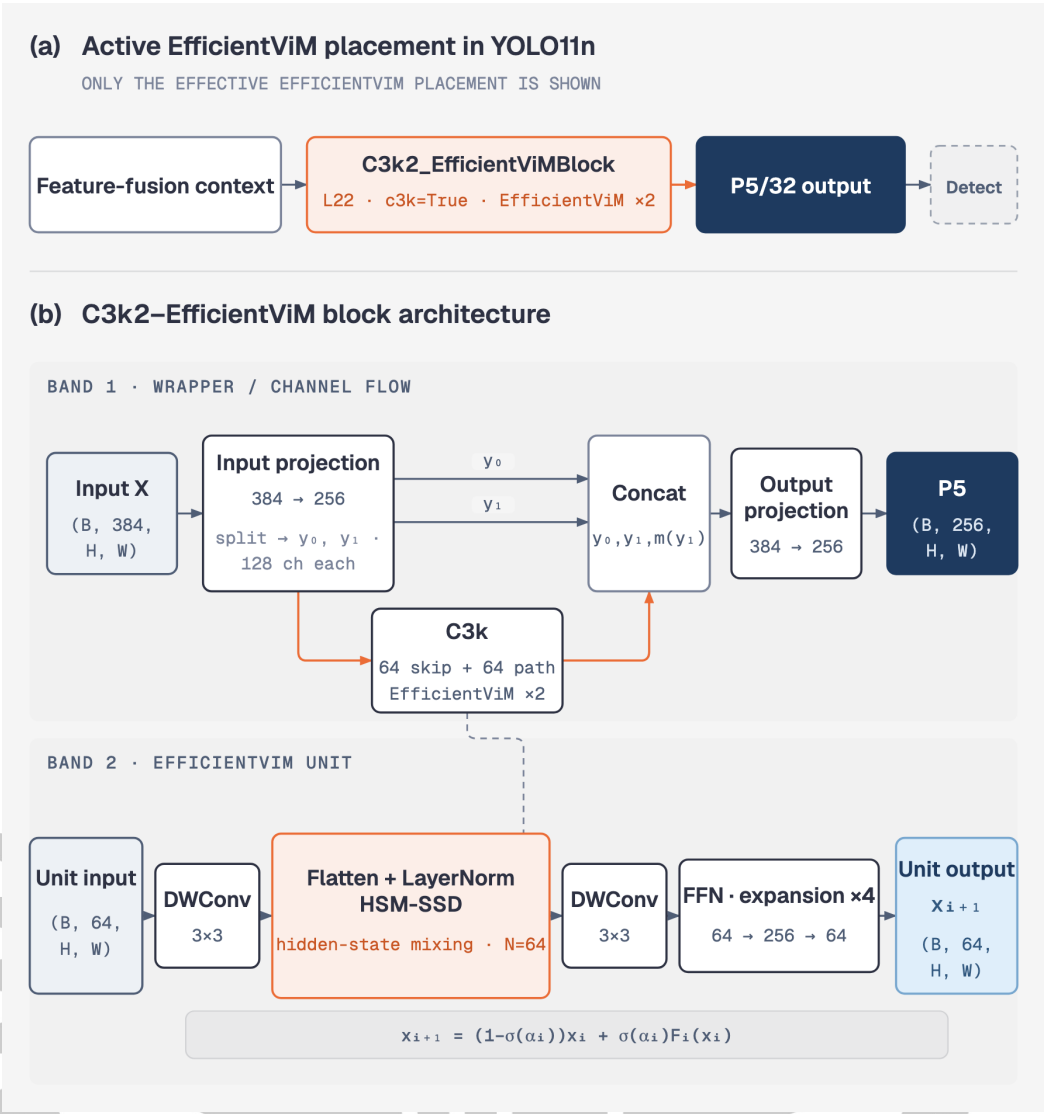


Figure 2: Author-generated schematic of the adapted EfficientViM Block. The upper panel marks its effective P5/32 neck placement, while the lower panels show the wrapper channel flow, four sub-operators, and gated residual in Equation (3.1).

3.3.2 DySample Dynamic Upsampling Module

DySample formulates dynamic upsampling as learned point sampling rather than kernel-based dynamic convolution [16]. It predicts offsets from a low-resolution feature map and uses standard bilinear grid sampling to construct the higher-resolution output. Let $X \in \mathbb{R}^{C\times h\times w}$ be the input, let s be the upsampling factor, and let p denote a location on the $sh \times sw$ output grid. Under the half-pixel convention, fixed bilinear interpolation maps p to the source coordinate

$$G(p) = \frac{p + 0.5}{s} - 0.5. \tag{3.2}$$

This geometry depends only on the output position and scale. DySample instead learns content-dependent displacements from this fixed bilinear sampling geometry.

The learnable offset field is

$$O = \lambda \left(W * X \right) \in \mathbb{R}^{2n_g s^2 \times h \times w}, \tag{3.3}$$

where W denotes the learnable weights of the 1×1 offset-prediction convolution, $*$ denotes convolution, n_g is the number of channel groups, and λ is a fixed scope factor. The code represents source coordinates in a pixel-edge coordinate system whose pixel centers are half-integers. For a low-resolution location (u, v), the source-center grid and displaced grid are

$$G_{\text{pixel}}(u, v) = (u + \tfrac{1}{2}, v + \tfrac{1}{2}), \tag{3.4}$$

$$Q = G_{\text{pixel}} + P_{\text{init}} + O, \tag{3.5}$$

where P_{init} contains the fixed bilinear sub-pixel offsets and Q contains the displaced source coordinates before normalization and rearrangement. Let Q_g denote the coordinate channels assigned to sampling group $g \in \{1, \ldots, n_g\}$. For $q = (q_x, q_y)$, the implementation uses

$$\text{norm}(q) = \left(\frac{2q_x}{w} - 1, \frac{2q_y}{h} - 1 \right),$$

which is the coordinate convention required by grid sampling with `align_corners=False`. The final sampling coordinate for group g at output position p is

$$S_g(p) = \text{PS}\big(\text{norm}(Q_g)\big)(p), \tag{3.6}$$

where norm maps source coordinates to the $[-1, 1]$ convention used by grid sampling and PS denotes pixel shuffle, which rearranges the s^2 sub-pixel coordinates to the higher-resolution output grid.

Equation (3.2) is the output-index form of the same half-pixel baseline represented in code by $G_{\text{pixel}} + P_{\text{init}}$ followed by pixel shuffle. For every low-resolution location and sub-pixel position, the two corresponding channels of O represent horizontal and vertical displacement. The C input channels are partitioned into n_g groups, and all channels within one group are resampled using that group's offset pair. Let $g(c) \in \{1, \ldots, n_g\}$ denote the sampling group assigned to input channel c. Thus, the geometry is learned per group rather than independently

for every channel. Because λ is fixed rather than predicted from X, this implementation uses the static-scope form of DySample.

The output feature is obtained by bilinear resampling at the coordinate of the group assigned to channel c,

$$X'(c,p) = \sum_{q \in \mathcal{N}(S_{g(c)}(p))} \kappa\big(q - S_{g(c)}(p)\big) X(c,q), \tag{3.7}$$

where $\mathcal{N}(S_{g(c)}(p))$ contains the four integer neighbours around the real-valued, group-specific source coordinate and $\kappa(u,v) = \max(0, 1-|u|)\max(0, 1-|v|)$ is the bilinear kernel. The executed `grid_sample` call uses bilinear interpolation, `align_corners=False`, and border padding. When $O = 0$, every S_g reduces to the corresponding bilinear-initialized grid.

The standalone variant replaces the two fixed nearest-neighbour upsampling operations in the baseline top-down neck: the P5-to-P4 transition and the subsequent P4-to-P3 transition. Both use $s = 2$, $n_g = 4$, $\lambda = 0.25$, low-resolution-first offset generation, and no dynamic scope modulation. For $s = 2$, P_{init} supplies the fixed ± 0.25 sub-pixel offsets, and the offset convolution produces $2n_g s^2 = 32$ channels. The conventional C3k2 blocks, bottom-up path, and detection head remain unchanged.

Figure 3 summarizes the two replacements and the learned sampling process.

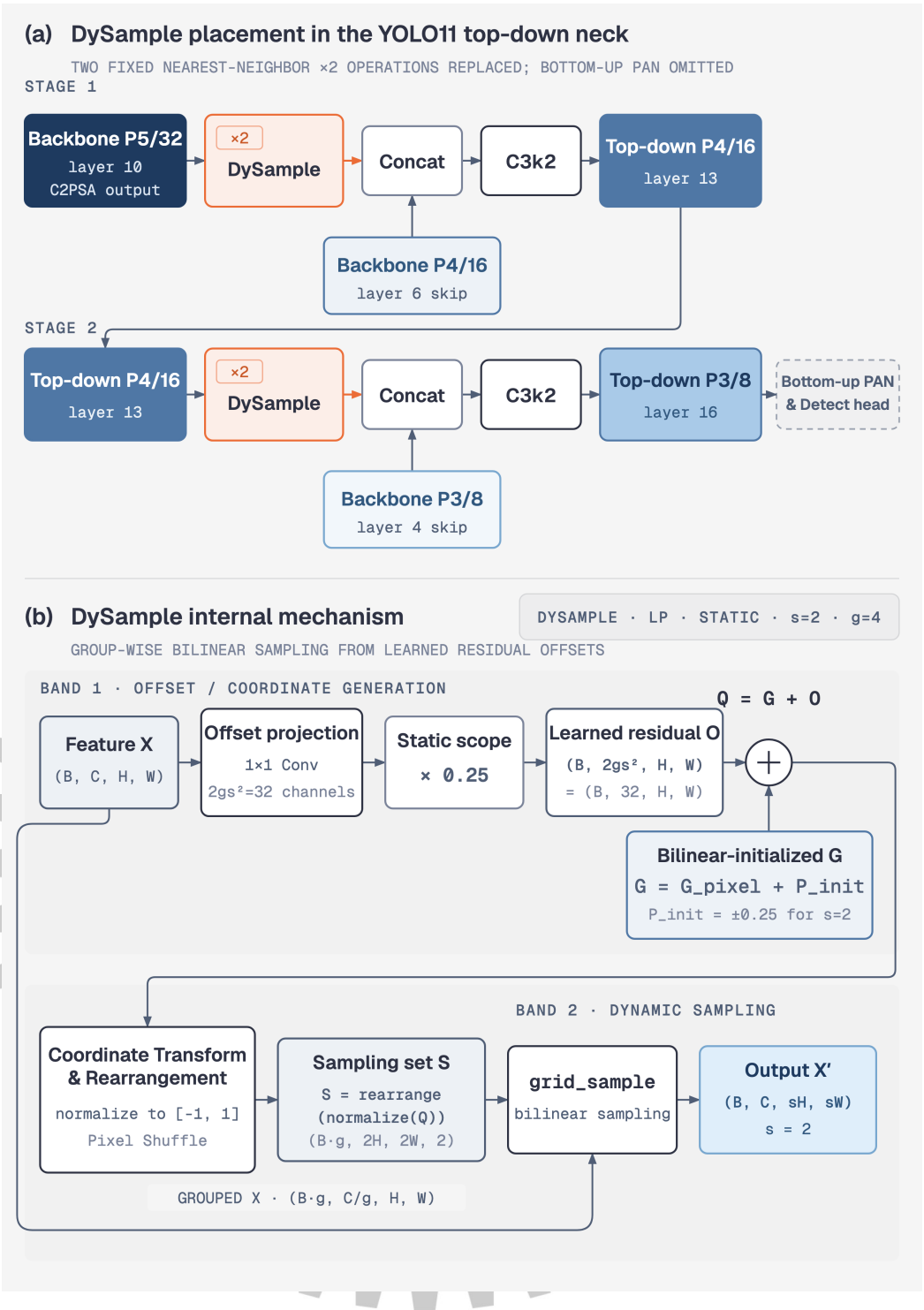


Figure 3: Author-generated schematic of the DySample variant. The upper panel marks the P5-to-P4 and P4-to-P3 replacements. The lower panel shows offset prediction, displacement of the bilinear-initialized grid, coordinate normalization and pixel shuffle, and standard bilinear grid sampling.

3.3.3 ASFF Multi-Scale Feature Fusion Head

ASFF aligns pyramid features to a common target level and learns a spatially adaptive contribution from each aligned source [17]. Following the source indexing, let $l \in \{1, 2, 3\}$ denote the target level, with level 1 corresponding to P5/32, level 2 to P4/16, and level 3 to P3/8. For source feature X^k, the alignment transform is

$$X^{k \to l} = \mathcal{T}_{k \to l}(X^k) \in \mathbb{R}^{C_l \times H_l \times W_l}. \tag{3.8}$$

$\mathcal{T}_{k \to l}$ is target-specific in the implementation. For target P5, P5 is unchanged, P4 uses a stride-two 3×3 convolution, and P3 uses stride-two max pooling followed by a stride-two 3×3 convolution. For target P4, P5 is first compressed by a 1×1 convolution and then upsampled twofold by nearest-neighbour interpolation, P4 is unchanged, and P3 uses a stride-two 3×3 convolution. For target P3, P5 and P4 are first compressed by 1×1 convolutions and then upsampled by factors four and two, respectively, while P3 is unchanged.

The weighting network does not predict each score map independently from one source. Instead, it compresses every aligned source to 16 channels, concatenates the three compressed tensors, and jointly predicts the three score maps:

$$\begin{aligned} Z^{k \to l} &= \mathcal{C}^{k \to l}_{1 \times 1}(X^{k \to l}) \in \mathbb{R}^{16 \times H_l \times W_l}, \\ \Lambda^l &= \mathcal{W}^l_{1 \times 1} * \mathrm{Cat}(Z^{1 \to l}, Z^{2 \to l}, Z^{3 \to l}) \in \mathbb{R}^{3 \times H_l \times W_l}. \end{aligned} \tag{3.9}$$

The three channels of Λ^l are denoted $\lambda^{1 \to l}$, $\lambda^{2 \to l}$, and $\lambda^{3 \to l}$.

At every position p, the three scores are normalized across source levels. Denoting the weights for sources 1, 2, and 3 by α^l, β^l, and γ^l, respectively,

$$\alpha^l(p) = \frac{\exp(\lambda^{1 \to l}(p))}{\sum_{j=1}^{3} \exp(\lambda^{j \to l}(p))}, \tag{3.10}$$

with β^l and γ^l defined identically from $\lambda^{2 \to l}$ and $\lambda^{3 \to l}$. The three non-negative weights sum to one at each position, and the fused feature is

$$\begin{aligned} \widetilde{Y}^{\,l}(p) &= \alpha^l(p) X^{1 \to l}(p) + \beta^l(p) X^{2 \to l}(p) + \gamma^l(p) X^{3 \to l}(p), \\ Y^l &= \mathcal{E}^l_{3 \times 3}(\widetilde{Y}^{\,l}). \end{aligned} \tag{3.11}$$

The alignment and weighting operations serve separate roles. $\mathcal{T}_{k\to l}$ first makes the three tensors compatible in spatial resolution and channel count; the joint predictor in Equation (3.9) then determines how much each compatible tensor contributes. Because the softmax is evaluated independently at every p along the three-source dimension, one image region can emphasize a high-resolution P3 contribution while another emphasizes the more abstract P5 contribution. The 3×3 expansion convolution then produces the target output sent to the detector. This spatial selection is the defining difference from assigning a single coefficient to each complete feature level.

The standalone ASFF variant retains the baseline neck and replaces the terminal `Detect` module with `DetectASFF`. Its three fusion paths align P3, P4, and P5 to each target resolution, jointly compute the three score maps, and form the expanded fused feature in Equation (3.11). The resulting P3, P4, and P5 tensors are then forwarded through the inherited `Detect` classification and box-regression branches. Thus, the terminal module is replaced, but the classification and regression routing remains after ASFF fusion. The weights are spatially varying rather than one global weight per feature level.

Figure 4 summarizes the three-level alignment and fusion process.

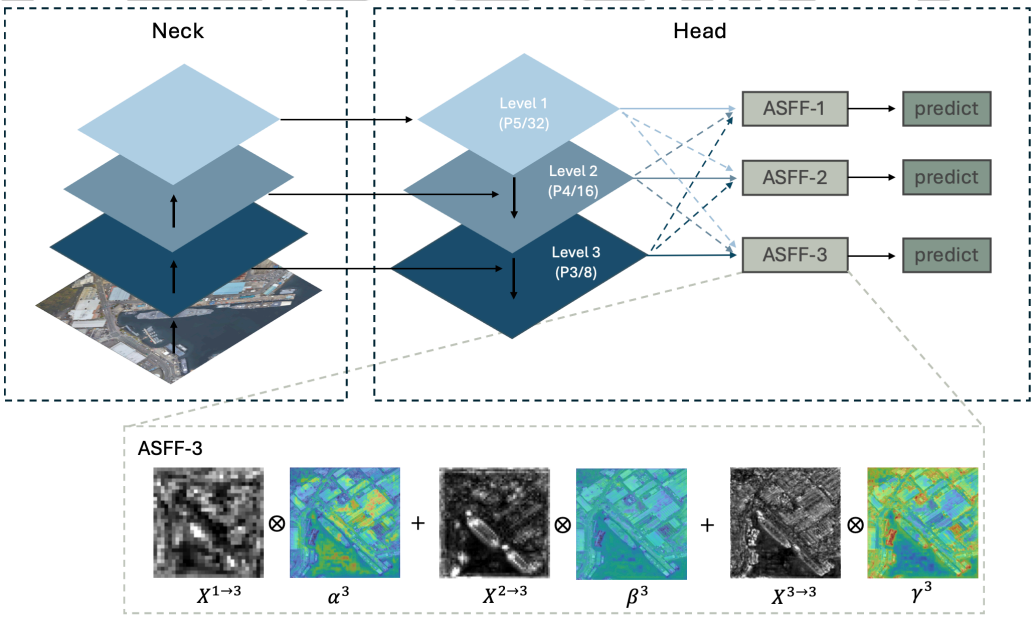


Figure 4: High-level author-generated schematic of the ASFF head. Level 1, Level 2, and Level 3 denote P5/32, P4/16, and P3/8. Every target branch receives all three levels after alignment and combines them with spatially varying softmax weights. For clarity, the diagram omits the 16-channel source compressions, concatenation, joint three-map predictor, and final expansion convolution defined in Equations (3.9)–(3.11).

3.4 Model Variant Configuration Summary

Table 2 consolidates the role and effective configuration of each standalone variant. Parameter counts are from the recorded four-class model instances. GFLOPs were profiled at a common 640×640 input and therefore are not operation counts for the 960×960 training input.

Table 2: Standalone module roles, insertion points, and recorded computational costs.

Model	Intervention	Inserted or replaced component	Effective location	Params (M)	GFLOPs
YOLO11n	Reference architecture	None	P3/P4/P5 detector	2.583	6.3
EfficientViM Block	Feature representation	EfficientViM-containing C3k branch	P5/32 neck output	2.560	7.6
DySample Module	Top-down sampling	Two fixed upsampling operations	P5-to-P4 and P4-to-P3	2.595	6.3
ASFF Head	Multi-scale fusion	Conventional terminal head	P3/P4/P5 prediction head	3.954	8.5

Figure 5 shows the three effective insertion locations in one diagram. The overlay marks where each intervention acts; each standalone variant activates only its own modification.

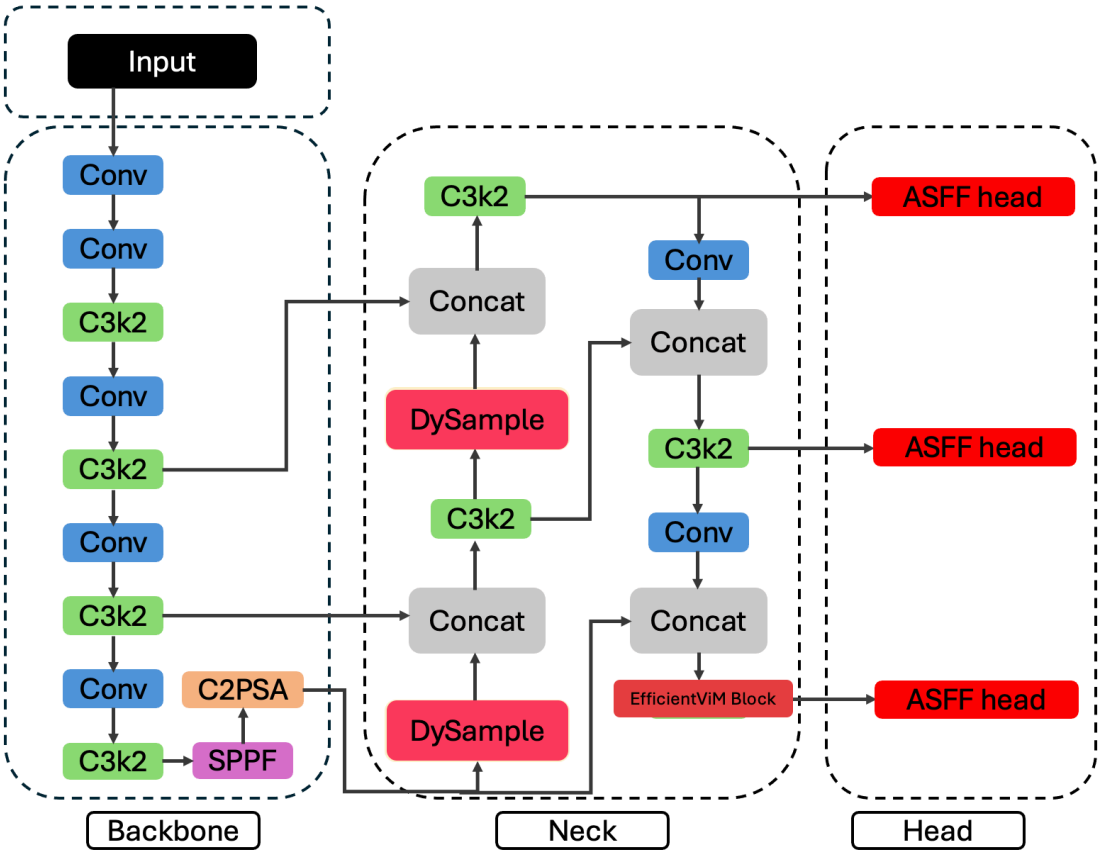


Figure 5: Author-generated schematic showing EfficientViM Block at the P5/32 neck output, DySample at the two top-down upsampling operations, and ASFF at the P3/P4/P5 prediction head.

Common training control. All retained core models use the same positive-sample assignment procedure and detection objective. Positive samples are selected by the Task-Aligned Assigner derived from Task Alignment Learning [23], which ranks candidate locations using both the predicted class response and ground-truth overlap. The assigner determines which locations and targets contribute to supervision; it is not an additional term in the loss function.

The box term $\mathcal{L}_{\mathrm{CIoU}}$ aggregates Complete Intersection over Union loss for overlap, center displacement, and aspect-ratio consistency [24]. The classification term $\mathcal{L}_{\mathrm{BCE}}$ is binary cross-entropy, while $\mathcal{L}_{\mathrm{DFL}}$ is Distribution Focal Loss for the discrete distributions encoding the four box-boundary distances [25]. With scalar gains λ_{box}, λ_{cls}, and λ_{dfl}, the common objective is

$$\mathcal{L} = \lambda_{\mathrm{box}}\mathcal{L}_{\mathrm{CIoU}} + \lambda_{\mathrm{cls}}\mathcal{L}_{\mathrm{BCE}} + \lambda_{\mathrm{dfl}}\mathcal{L}_{\mathrm{DFL}}. \tag{3.12}$$

The fixed gains are $(\lambda_{\mathrm{box}}, \lambda_{\mathrm{cls}}, \lambda_{\mathrm{dfl}}) = (7.5, 0.5, 1.5)$. They are the loss-gain settings in the version-pinned Ultralytics training configuration used by this study [8], not quantities derived

from the component-loss equations or parameters learned during training. The software version and remaining execution settings are reported in Section 5.1.

All variants retain this assignment procedure, these loss components, and their gains; only the intervention summarized in Table 2 changes. Using the same optimization target controls for differences in supervision across the architecture comparison. The variants are evaluated in terms of task-specific accuracy and computational cost rather than ranked by a single universal criterion.

3.5 Foreground Compositing and Annotation Generation

The data-side method applies one general pipeline for compositing RGBA foregrounds and generating their annotations. The shared operations are foreground loading, geometric and appearance transformation, alpha compositing, tight-box derivation from the transformed alpha mask, class assignment, preservation of host annotations, and operator review. V1 uses this pipeline for two distinct applications: adding harbor vehicles and constructing the airplane–vessel co-occurrence missing from the source harbor imagery. Foreground and host suitability, semantic placement, transformation choices, class assignment, and final acceptance remain human decisions; the software implementation is described in Chapter 4.

3.5.1 Foreground Transformation and Placement Constraints

Let the host image occupy the discrete domain

$$\Omega = \{0, 1, \ldots, W-1\} \times \{0, 1, \ldots, H-1\}, \tag{3.13}$$

where W and H are its width and height. The operator defines the Smart Zone as the axis-aligned rectangle

$$\mathcal{Z} = \{(x, y) \in \Omega : z^x_{\min} \leq x \leq z^x_{\max},\ z^y_{\min} \leq y \leq z^y_{\max}\}. \tag{3.14}$$

Let T_θ denote the operator-selected transformation, including placement, scale, rotation, and any appearance adjustment applied to the RGBA foreground. After transformation, its visible

support induces the half-open bounding box

$$\mathcal{B} = \{(x, y) \in \Omega : b^x_{\min} \le x < b^x_{\max},\ b^y_{\min} \le y < b^y_{\max}\}. \tag{3.15}$$

The software accepts a Smart Zone placement only when

$$\mathcal{B} \subseteq \mathcal{Z} \iff z^x_{\min} \le b^x_{\min} \ \wedge\ b^x_{\max} \le z^x_{\max} + 1 \ \wedge\ z^y_{\min} \le b^y_{\min} \ \wedge\ b^y_{\max} \le z^y_{\max} + 1. \tag{3.16}$$

The Smart Zone is an optional geometric constraint defined by the operator. Manual mode permits placement without the containment test; when Smart Zone mode is enabled, the software applies Equation (3.16). Neither mode detects a flight deck, infers a support surface, or assesses the semantic plausibility of the selected region. For harbor-vehicle compositing, the operator reviews whether the transformed foreground is plausibly situated in the visible harbor scene. For shipboard-aircraft context construction, the operator instead assesses whether the selected host and placement form a plausible airplane–vessel or deck relationship. The available records do not establish that Smart Zone mode was enabled for every instance; the containment condition therefore describes the enabled placement mode rather than a universal property of all composites.

3.5.2 Compositing and Synchronized Annotation Generation

Let $F = T_\theta(A)$ be a transformed RGBA foreground asset, let F_{rgb} be its color channels, let $\alpha \in [0, 1]$ be its transformed alpha channel, and let B be the host image. At every host coordinate $(x, y) \in \Omega$, the composited image is

$$I(x, y) = \alpha(x, y)F_{\mathrm{rgb}}(x, y) + \big(1 - \alpha(x, y)\big)B(x, y). \tag{3.17}$$

The annotation is derived from the same transformed alpha channel. Its visible support is

$$\mathcal{S} = \{(x, y) \in \Omega : \alpha(x, y) > \tau\}, \qquad \tau = \frac{16}{255}, \tag{3.18}$$

where 16 is the strict threshold used by the audited current implementation on its 8-bit alpha mask. For non-empty $\mathcal{S}$, the tight axis-aligned box is

$$b^x_{\min} = \min_{(x,y)\in\mathcal{S}} x, \qquad b^x_{\max} = 1 + \max_{(x,y)\in\mathcal{S}} x, \tag{3.19}$$

$$b^y_{\min} = \min_{(x,y)\in\mathcal{S}} y, \qquad b^y_{\max} = 1 + \max_{(x,y)\in\mathcal{S}} y. \tag{3.20}$$

Computing the box from $\mathcal{S}$ rather than from the rectangular RGBA canvas excludes transparent or near-transparent margins introduced by rotation, interpolation, or the original cut-out boundary. The threshold is fixed in the current implementation. The available V1 records provide aggregate added-box counts but not the exact generating revision or a per-instance transformation log; the threshold and Smart Zone mode therefore cannot be confirmed retrospectively for every composite.

After clipping the box to the host extent, the added object is written in normalized YOLO form,

$$x_c = \frac{b^x_{\min} + b^x_{\max}}{2W}, \qquad y_c = \frac{b^y_{\min} + b^y_{\max}}{2H}, \tag{3.21}$$

$$w = \frac{b^x_{\max} - b^x_{\min}}{W}, \qquad h = \frac{b^y_{\max} - b^y_{\min}}{H}. \tag{3.22}$$

An empty or degenerate clipped support is rejected. Otherwise, the operator-selected class record is appended while the host annotations are preserved, and the completed image and label file are saved together. Deriving both the composited foreground and its tight box from the same transformed alpha channel and committed placement state maintains image–annotation synchronization, while the explicit threshold defines which alpha pixels contribute to the box.

At export, harbor-vehicle composites are assigned to the large vehicle or small vehicle class, whereas shipboard-aircraft composites are assigned to the airplane class. Both applications use the same transformation, alpha-compositing, box-generation, and image–label export operations. Their class choices and semantic placement criteria differ, but the synchronized annotation-generation rule remains unchanged. Chapter 5 reports their aggregate training counts and their joint role in the V1 treatment.

Chapter 4 illustrates how the application operationalizes this sequence, including transformation preview and final image–label inspection. Chapter 5 reports the versioned datasets and experimental conditions in which the resulting image–label pairs are used.

Chapter 4. Software Tools for Data Preparation and Post-Training Detection Analysis

4.1 Data-to-Analysis Workflow and Tool Roles

The three software tools in this chapter serve different stages of the research workflow. Before training, the GUI-based dataset tool is used to inspect data, review annotations, perform foreground compositing, and synchronize labels. The multi-view vessel-type classifier helps the researcher review uncertain vessel annotations. After training, the web-based platform runs selected checkpoints and presents predictions, summary statistics, stored results, auxiliary interpretations, and exportable reports. Their relationship to the study is summarized as

$$\text{Data preparation tools} \longrightarrow \text{Training} \longrightarrow \text{Detection-analysis platform.}$$

These tools provide research infrastructure rather than independent detection methods. The GUI makes data construction inspectable and repeatable, the vessel-type classifier assists annotation correction, and the web platform links post-training inference to stored results. Section 3.5 defines Smart Zone placement, alpha compositing, and label synchronization; this chapter explains how the software applies those procedures. Chapter 5 evaluates detector performance and classifier assistance separately.

All three tools preserve a clear division between software operation and human judgement. The software parses annotations, enforces geometric constraints, ranks candidate vessel types, runs a selected checkpoint, and stores results. The researcher remains responsible for judging the plausibility of a constructed scene, accepting or rejecting an assisted label, and interpreting

detector predictions or generated explanations.

4.2 GUI-Based Dataset Annotation and Foreground Compositing

The dataset application provides an interactive workspace for preparing the study data. It imports images and annotations, summarizes dataset content, and displays bounding boxes so that missing, inconsistent, or questionable labels can be reviewed before training. Dataset review, image editing, and annotation handling are thus carried out within one visual workflow.

For foreground compositing, the application loads pre-existing RGBA assets and preserves their alpha channels during transformation and preview. The operator selects a harbor host and a suitable foreground, then adjusts its placement and appearance while viewing the proposed composite. Manual mode permits direct placement. In Smart Zone mode, the operator draws an allowable rectangular region, and the software applies the containment check defined in Section 3.5.1. This check enforces a geometric constraint; identifying a flight deck and assessing physical plausibility remain human tasks.

When the operator accepts a placement, the application composites the transformed foreground and derives its bounding box from the transformed alpha mask using the strict threshold specified in Section 3.5.2. It writes the new object annotation with the image while preserving the existing host annotations. Because the box is derived from thresholded alpha support, low-alpha edge pixels are not automatically included. A final preview lets the researcher inspect the image, class, and box before accepting or rejecting the constructed example.

Table 3 summarizes the resulting division of responsibility. The application reduces clerical and geometric error, but host suitability, foreground suitability, transformation choices, and final acceptance remain human decisions.

Table 3: Dataset-system functions and human responsibility in the research workflow.

Research step	Software support	Human responsibility
Dataset inspection	Parse images and labels; display boxes and counts	Select the intended dataset and review anomalies
Foreground preparation	Load and preview RGBA assets	Approve asset and host suitability
Placement	Provide manual controls and Smart Zone containment	Draw the allowable region and choose the placement
Transformation preview	Apply and preview appearance transformations	Judge local visual consistency
Annotation synchronization	Derive and export the composited-object box	Confirm the class and inspect the final box
Quality control	Present the completed image–label pair	Accept, revise, or reject the example

Figure 6 shows the main research workspace. It combines image and annotation inspection

with foreground selection, transformation preview, and synchronized export; Smart Zone mode adds the user-defined placement constraint to the same workflow.

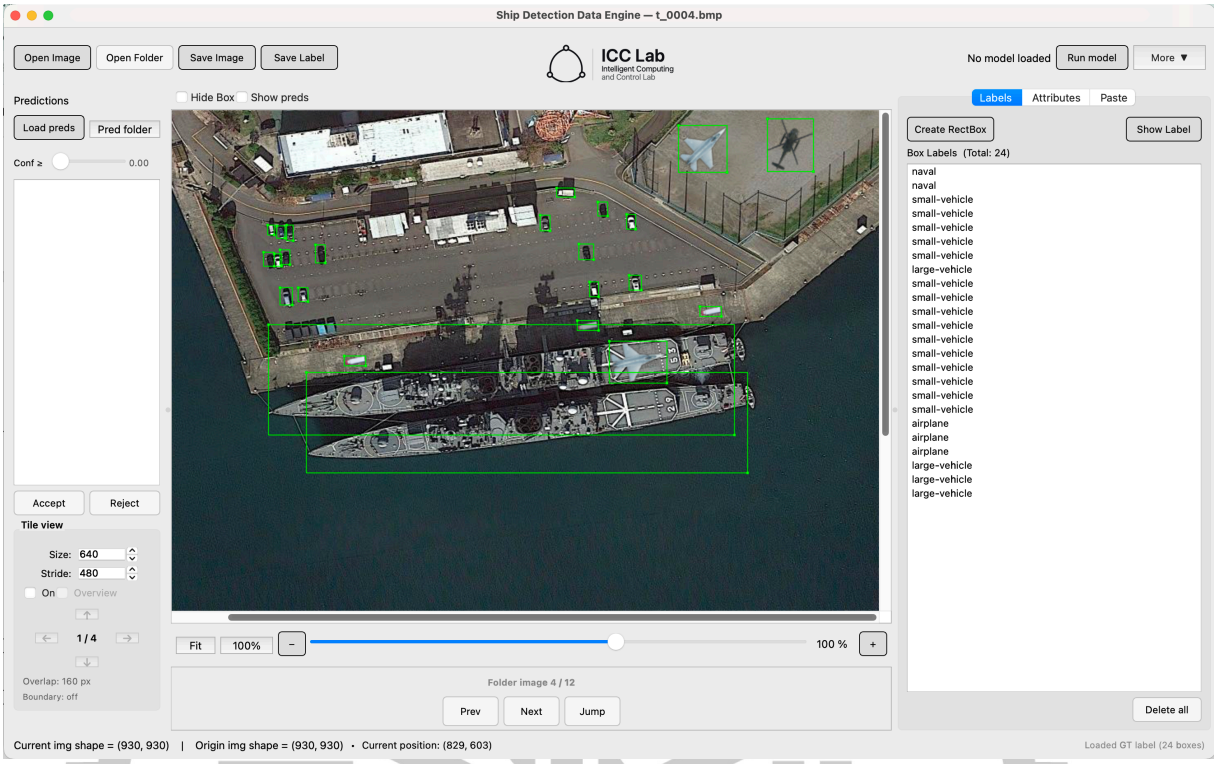


Figure 6: Main workspace for inspecting images and annotations, previewing transformed RGBA foregrounds, and exporting accepted image–label pairs.

Figure 7 summarizes the operator-facing sequence across four stages. The researcher first loads a source image, selects an RGBA foreground, and configures its scale, rotation, and appearance. The transformed foreground is then positioned through a live preview; in Smart Zone mode, the containment test in Equation (3.16) constrains this placement before confirmation. Once the preview is accepted, the application saves the composite image and its matching bounding-box labels from the same committed placement state. The workflow makes synchronized image–label export an explicit final operation rather than an independent annotation step.

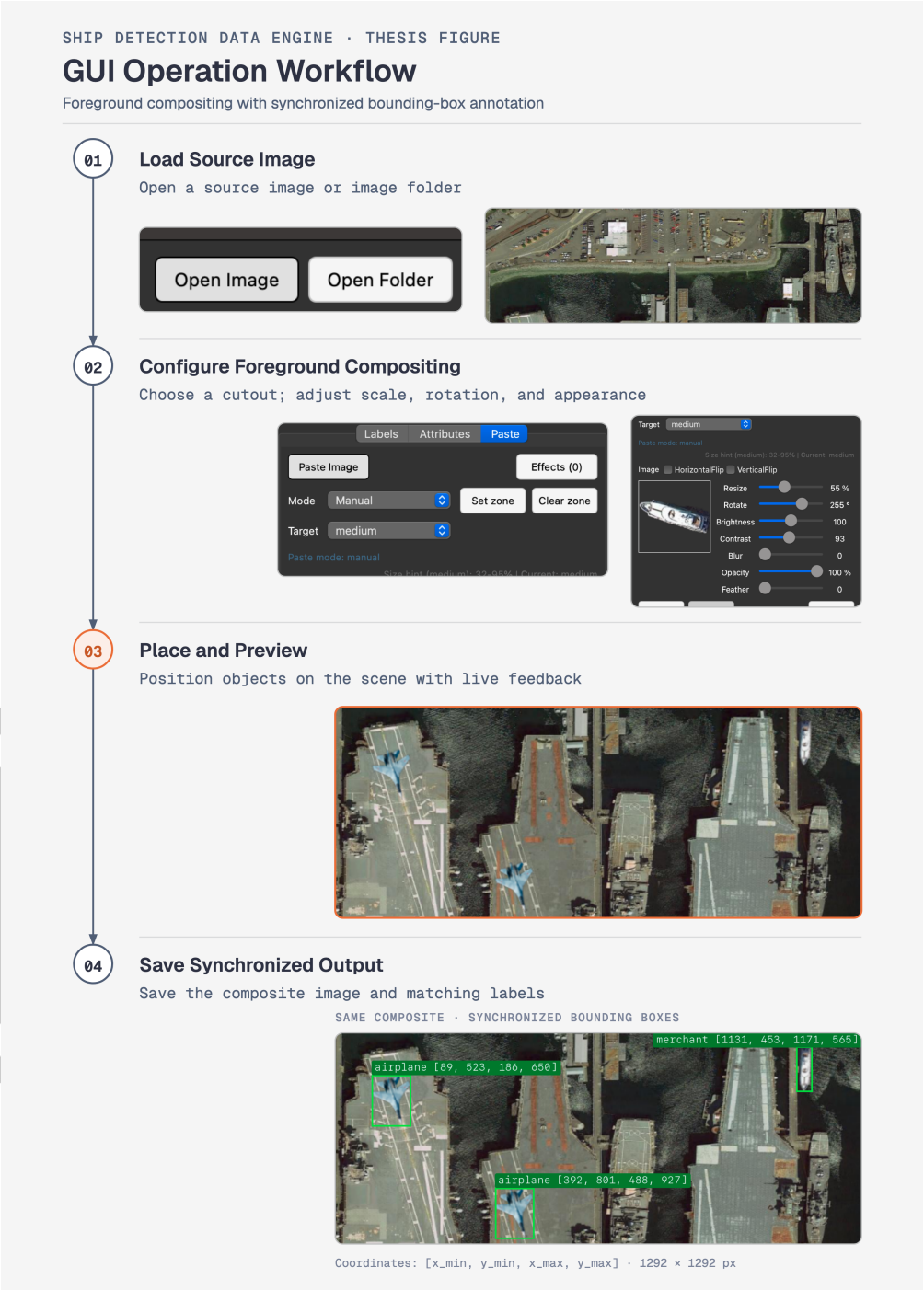


Figure 7: Operator-facing workflow of the GUI-based dataset annotation and foreground-compositing tool. A source image and RGBA foreground are loaded, transformed, and previewed; confirmation writes the composite image and matching bounding-box labels from the same accepted placement.

4.3 Multi-View Vessel-Type Classifier for Annotation Assistance

The Multi-View Vessel-Type Classifier is a pre-training aid for cases in which a researcher has already identified a missing or questionable vessel annotation. Given a source image and a researcher-supplied vessel box, it ranks the candidate classes `naval`, `merchant`, and `other_vessel`. The `dock` class is excluded because it represents fixed infrastructure rather than a vessel type. The tool performs classification rather than localization, and any label change requires researcher confirmation.

This supplied-box design matches the annotation workflow. The researcher first identifies a vessel that is missing from the labels or has an uncertain type, then defines the region to be reviewed. The box records which object requires attention, while the assistant addresses only its vessel type. Any proposed class is conditional on the supplied region.

The assistant uses two complementary views of the supplied region. A tight crop emphasizes the vessel, while a context crop preserves surrounding information that may help distinguish vessels with similar overhead appearance. A frozen multi-view ensemble combines features from both views and incorporates coarse vessel classes and finer subtype structure before producing the three-class ranking. The ensemble is designed to assist review rather than label data autonomously. Appendix A.6 records the crop, ensemble, bias, specialist-rule, checkpoint, and artifact settings.

The two views also provide a practical basis for review. The tight crop shows whether the supplied box captures the vessel adequately, while the context crop exposes nearby structures without changing the box. Combining coarse-class and subtype-informed features reduces dependence on a single representation. The frozen ensemble also ensures that repeated use of the same input follows the same decision policy, although the researcher must still judge whether the proposed class is correct.

The resulting scores are temperature-calibrated before presentation so that the displayed probabilities are more suitable for comparing and ranking candidates. The interface presents the ranked list with both image views. The researcher may accept a candidate, select another class, revise the supplied region, or reject the proposal. The software narrows the classification task after a vessel has been identified, while the semantic annotation decision remains human.

An accepted proposal returns to the annotation workflow for final inspection with the image

and box. Rejection leaves the source annotation unchanged and allows the researcher to reconsider the region or assign a class manually. In this way, the module reduces an open-ended typing decision to a short ranked list without removing the final human acceptance step.

Figure 8 shows the corresponding GUI sequence. The researcher loads the source image, frames the target vessel, and may refine its segmentation with positive or negative prompts. The retained workflow records identify this segmentation stage as the original Segment Anything Model (SAM), not SAM2, using the supplied box with optional positive or negative point prompts [26]. The software creates the cutout, forms the two classification views, and presents a vessel-type proposal with the mask, class scores, and confidence cues. If either the mask or class is unacceptable, the workflow returns to segmentation refinement. Chapter 5 evaluates the usefulness of this assistance under different input conditions.



Figure 8: Operator-facing workflow of the multi-view vessel-type classifier. The researcher identifies and refines the vessel region, the software generates a two-view class proposal, and the researcher accepts the result or returns to segmentation refinement.

4.4 Web-Based Detector Inference and Result-Inspection Platform

The YOLO-based Object Detection and Intelligent Analysis Platform supports the post-training workflow; model training takes place separately. For analysis, the researcher selects a registered detector, supplies a still image, and sets the confidence and non-maximum-

suppression IoU thresholds. Registration assigns each inference an explicit model identity and class mapping and prevents an unrecorded checkpoint from being selected. Appendix A.5 summarizes the registration logic, stored result fields, and reproducibility limits.

After validating the request, the platform runs the selected checkpoint and stores the result with the input image, model identity, class mapping, thresholds, predicted classes, confidence values, and bounding boxes. This association allows an inference to be revisited with its model and settings intact. The workflow evaluated in this thesis is limited to uploaded still images and the registered horizontal-bounding-box detectors described in the appendix.

The result view overlays the predicted boxes and class labels on the input and presents structured object rows, class counts, and a history of stored inferences. Confidence and IoU controls allow the researcher to inspect how the visible prediction set changes under different operating settings. These outputs support result inspection; they remain detector predictions rather than annotations or evaluation metrics. Figure 9 shows how the history and the selected provenance-bearing record are reviewed together.

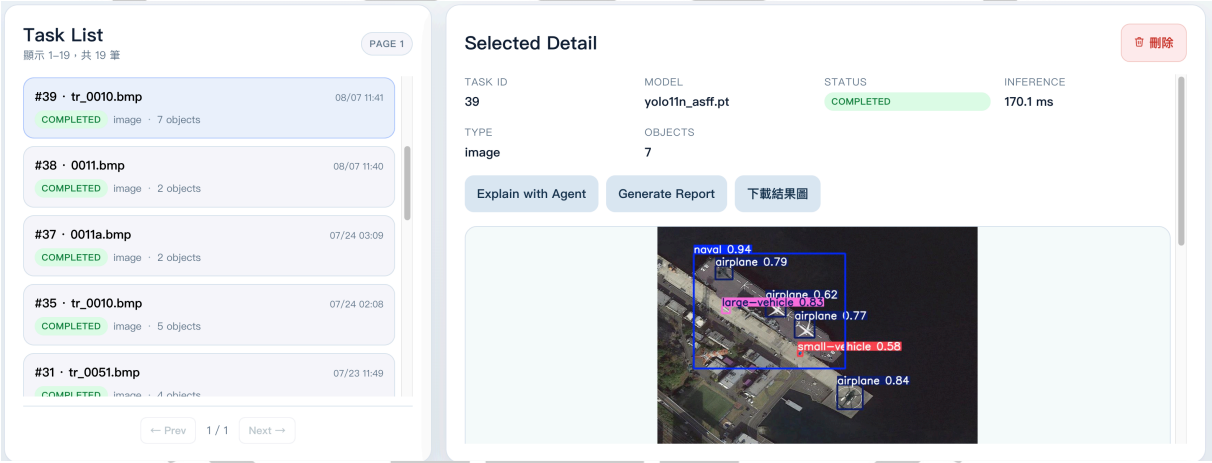


Figure 9: Interface for reviewing inference history and a selected result. The displayed bounding boxes, class labels, confidence values, and call time describe one stored inference; they are not measures of detector accuracy or throughput.

Stored results can be exported as Markdown or PDF reports. An optional AI-assisted layer converts selected result fields into explanatory prose, but it operates after detection and cannot alter boxes, classes, confidence values, or provenance. In light of the reporting and generative-AI risks discussed in the literature [27, 28], this text is treated as an auxiliary interpretation that requires human review. It does not serve as annotation ground truth, expert judgement, or independent validation.

Figure 10 summarizes the post-training data flow. Detector output and stored metadata

provide the source for visualization, history, auxiliary interpretation, and report export.

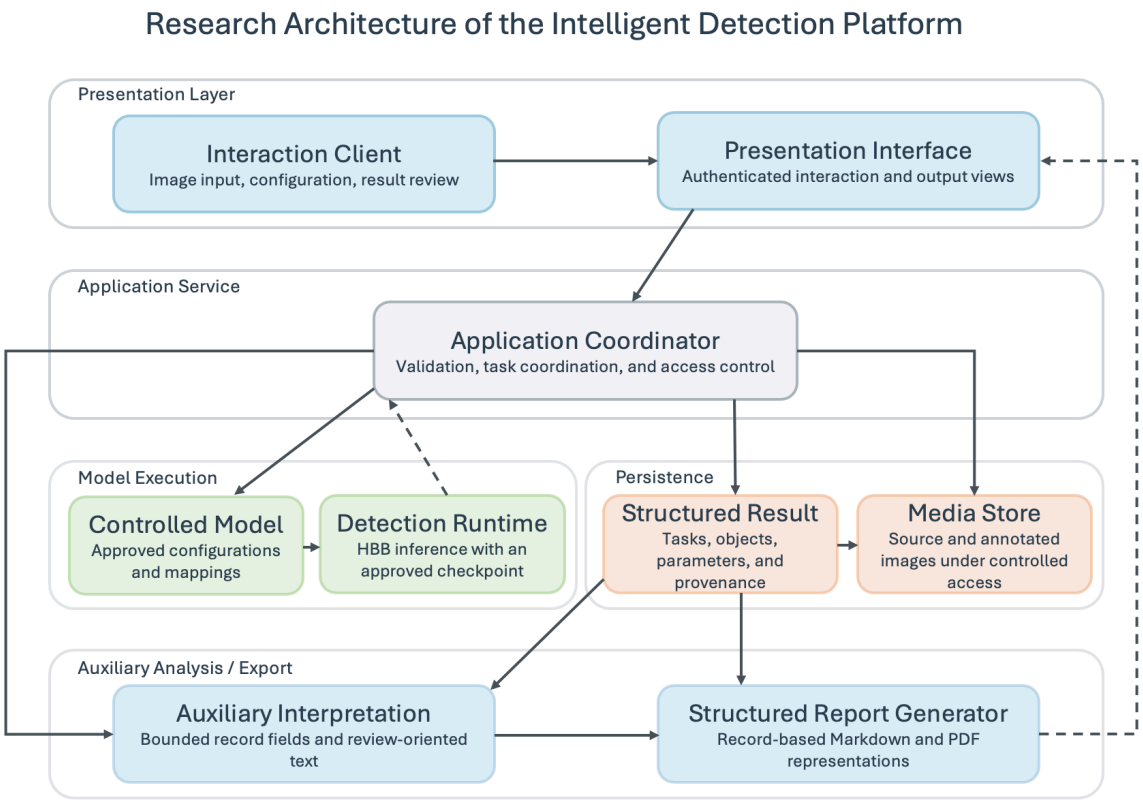


Figure 10: Post-training research flow from registered checkpoint inference to a provenance-bearing result record, visualization, history, auxiliary interpretation, and report export.

4.5 Research Roles, Functional Scope, and Verification

Table 4 summarizes how each tool contributes to the research workflow and how its functions were checked. *Verified* functions were executed and recorded. *Implemented/source-supported* functions were confirmed in the software or interface, although no complete formal test matrix is available. *Not evaluated* identifies operational properties that were outside the study.

Table 4: Software-tool roles, verification status, and research scope.

Tool or property	Verification status	Role in the research workflow
Dataset Annotation and Compositing GUI	Implemented/source-supported	Dataset and annotation inspection, RGBA handling, manual and Smart Zone placement, transformation preview, synchronized labels, and human review; a complete formal test matrix is unavailable.
Multi-View Vessel-Type Classifier	Verified workflow and frozen configuration	Supplied-box, calibrated candidate rankings for human annotation correction; Chapter 5 evaluates their usefulness.
Detector Inference and Result-Inspection Platform	Verified local workflow	Registered-model still-image inference, stored results with model provenance, visualization and history, auxiliary interpretation, and Markdown/PDF export.
Deployment and operational validity	Not evaluated	Production deployment, concurrency, complete security, fixed-hardware real-time latency, expert report quality, and operational validity lie outside this study.

The two pre-training tools provide an inspectable path from source data to accepted image–label pairs, while the post-training platform links a selected checkpoint and input to stored predictions and reports. These workflows organize and document the researcher's decisions; they do not establish detector correctness or replace semantic judgement. Appendices A.5 and A.6 summarize the implementation scope, provenance controls, and frozen assistant configuration.

Chapter 5. Experiments and Results

This chapter follows the progression of the research. YOLO11n architecture comparisons first revealed class- and metric-dependent trade-offs. Detector-error inspection then identified missing and inconsistent vessel annotations, leading to the controlled comparison of training-label states under Conditions A and B. The corrected harbor vessel labels were subsequently integrated with the airport-derived classes to form a seven-class harbor–airport setting. In this setting, V0 is the corrected baseline, and V1 adds manual harbor-vehicle annotations, foreground-composited harbor vehicles, and foreground-composited shipboard aircraft to the same training dataset. The V0/V1 pair evaluates this single harbor-domain supervision treatment. Appendix A contains the detailed provenance registries, calibration settings, checkpoint identities, and supplementary matrices.

5.1 Datasets, Conditions, and Training Settings

The experiments use a common YOLO11n framework at an input size of 960×960 pixels. Unless a retained historical record is explicitly identified, training uses 150 epochs, batch size 16, patience 100, automatic optimizer selection, deterministic execution, and seed 0. Each distinct trained model is represented by one run, except for the separate three-seed baseline variability record retained in the appendix. The controlled comparisons use identical evaluation membership within each pair; evaluation labels are never changed between the two members of a formal pair.

Table 5 summarizes the execution environment. Appendix A.4 provides a compact registry of the formal experiments and the reconciled historical runs. Full package captures, hashes, and machine-readable manifests remain in the external experiment archive.

The YOLO11 version 11.0.0 in the software citation identifies the cited 2024 model release [8]; Ultralytics 8.3.242 in Table 5 is the installed Python package version used for the recorded experiments. These identifiers describe different layers of provenance and are not contradictory.

The requested optimizer setting was `auto`. Execution logs resolve the selected optimizer

Table 5: Common hardware and software environment for the controlled detector experiments.

Item	Recorded value
GPU	NVIDIA GeForce RTX 4090, 24,564 MiB
CPU and memory	Intel Core i7-14700, 62 GiB RAM
Operating system	Ubuntu 24.04.2 LTS
Python	3.10.18
Ultralytics	8.3.242
PyTorch / torchvision	2.1.2+cu118 / 0.16.2+cu118
Training precision / device	Automatic mixed precision / CUDA device 0

for some, but not all, historical runs, so identical resolved optimizer parameters are not assumed when the terminal log is unavailable. Base online augmentation used mosaic 1.0, horizontal flip 0.5, translation 0.1, and scale 0.5, with mixup disabled. Foreground compositing was encoded only in the frozen training data and was not applied online. Formal evaluation used confidence 0.001, non-maximum-suppression IoU 0.7, and at most 300 detections per image.

AP@50 and AP@75 denote class average precision at IoU thresholds 0.50 and 0.75, respectively, while AP@50–95 averages the ten IoU thresholds from 0.50 through 0.95. mAP is the unweighted mean over the classes in the stated task. Precision and Recall, where reported for retained earlier experiments, are read at the confidence that maximizes F1. Consequently, AP describes the ranked precision–recall curve, whereas Precision and Recall describe one operating point. Changes are reported in percentage points when the tables use the 0–100 scale.

Confusion matrices use a separate fixed operating point. In the recorded Ultralytics 8.3.242 validation path, predictions are first filtered by NMS using the reported IoU 0.7 and at most 300 detections per image. When the default validation confidence 0.001 is passed to the detection confusion-matrix routine, that routine applies an effective confidence threshold of 0.25 and matches boxes at IoU 0.45. This behavior is independently count-verified for the retained Condition B DySample matrix. The other copied Condition A/B and harbor V0/V1 matrix images do not retain per-file arguments or prediction metadata, so their operating point cannot be independently recovered file by file; they are interpreted only as fixed-artifact error counts and not as AP evaluated at confidence 0.001.

Secondary metric availability differs across the retained experiment families. The single-class record reports AP@50 and AP@50–95; the Condition A and A/B records retain AP@50, AP@75, and AP@50–95; and the final V0/V1 records retain AP@50 and AP@50–95. AP@75 is used below as an additional localization-specific metric where it is available. Unavailable metrics are not reconstructed for cosmetic uniformity.

The original harbor source contains 3,435 images and the four classes `naval`, `merchant`, `dock`, and `other_vessel`. Table 6 reports a direct recount from the original source directories. These source-label statistics are distinct from the later corrected vessel labels and V1 additions.

Table 6: Original harbor source-dataset statistics recounted from the final source directories.

Dataset	Images	Class	Instances	Object Scale		
				Small	Medium	Large
Training Set	2,198	Naval	4,583	75	709	3,799
		Merchant	4,387	1,376	1,781	1,230
		Dock	744	0	28	716
		Other Vessel	1,146	236	428	482
Validation Set	550	Naval	1,156	7	182	967
		Merchant	1,473	751	391	331
		Dock	154	0	8	146
		Other Vessel	320	71	133	116
Test Set	687	Naval	1,441	14	165	1,262
		Merchant	1,601	719	497	385
		Dock	195	0	12	183
		Other Vessel	373	83	140	150

The filtered airport source contains 2,320 images: 1,662 training, 603 validation, and 55 test images. Table 7 reports the corresponding direct recount from the current source files.

Table 7: Final airport source-dataset statistics used by the completed seven-class experiments.

Dataset	Images	Class	Instances	Object Scale		
				Small	Medium	Large
Training Set	1,662	Airplane	4,744	293	3,714	737
		Large Vehicle	5,602	3,815	1,769	18
		Small Vehicle	64,325	62,367	1,958	0
Validation Set	603	Airplane	3,095	124	2,901	70
		Large Vehicle	1,953	1,707	246	0
		Small Vehicle	48,462	47,758	704	0
Test Set	400	Airplane	937	298	465	174
		Large Vehicle	1634	494	1094	46
		Small Vehicle	4876	4172	704	0

For both tables, object scale follows the COCO bounding-box area thresholds [29] in original source-image pixels: Small when $A < 32^2$, Medium when $32^2 \leq A < 96^2$, and Large when $A \geq 96^2$. Normalized YOLO HBB widths and heights are converted with the matched image's original width and height; the 960×960 training resize is not used. Airport test labels stored as pixel `xyxy` coordinates are handled in their original image coordinate space. The three airport-derived classes are documented in [2, 12], and the harbor source is documented in [1]. Their joint mapping is shown in Table 8.

Table 9 defines the final condition lineage. Old GT is historical pre-audit evidence, not a

Table 8: Canonical class mapping used in the seven-class experiments.

Joint ID	Class	Source group
0	`naval`	Harbor
1	`merchant`	Harbor
2	`dock`	Harbor
3	`other_vessel`	Harbor
4	`airplane`	Airport / constructed harbor
5	`large-vehicle`	Airport / constructed harbor
6	`small-vehicle`	Airport / constructed harbor

member of the Condition A/B treatment pair. Condition A and Condition B share the same corrected and frozen evaluation labels and differ only in the vessel training-label state. V0 and V1 form the only formal seven-class treatment pair.

Table 9: Dataset conditions and their experimental roles.

Condition	Training labels / data	Validation and test reference	Role
Old GT	Original vessel labels	Original labels	Historical pre-audit evidence
Condition A	Original vessel training labels	Corrected and frozen vessel evaluation labels	Annotation-quality control
Condition B	Partially corrected, human-confirmed vessel training labels	Exactly the same corrected and frozen evaluation labels as A	Training-label treatment
V0	Corrected harbor vessel labels plus airport-derived data; no added harbor vehicle supervision and no added shipboard-aircraft training samples	Corrected/frozen seven-class evaluation records	Corrected seven-class baseline
V1	V0 plus manual harbor-vehicle annotations, foreground-composited harbor vehicles, and foreground-composited shipboard aircraft in one training dataset	Same evaluation populations as V0 within each retained comparison	Combined targeted harbor-domain supervision treatment

Formal results are interpreted only within the controlled pairs Condition A versus Condition B and V0 versus V1. The A/B experiment concerns training-label quality. The V0/V1 experiment concerns the combined V1 supervision package; its three components are not separate scientific conditions.

5.2 YOLO11 Module Comparison

This section asks whether EfficientViM Block, DySample, ASFF, or their combinations provide a consistently superior architecture. The preliminary single-class experiment provides screening evidence, and the four-class comparison uses Condition A as the corrected and frozen evaluation reference.

5.2.1 Preliminary Single-Class Comparison

Table 10 retains the complete eight-model single-class screening record. These are historical single-run results on the FP-relabeled vessel setting, so they are used to compare patterns rather than to establish statistical superiority.

Table 10: Preliminary single-class vessel comparison. Values are on a 0–1 scale.

Model	Precision	Recall	AP@50	AP@50–95
YOLO11n	0.902	0.858	0.907	0.729
+ EfficientViM Block	0.905	0.845	0.905	0.719
+ DySample	0.916	0.847	0.907	0.722
+ ASFF	0.911	0.841	0.900	0.721
+ EfficientViM Block + DySample	0.920	0.836	0.909	0.723
+ EfficientViM Block + ASFF	0.906	0.835	0.900	0.718
+ DySample + ASFF	0.896	0.850	0.907	0.722
+ EfficientViM Block + DySample + ASFF	0.919	0.837	0.904	0.723

Different metrics favor different configurations. The EfficientViM–DySample combination has the highest Precision and AP@50, while the unmodified YOLO11n baseline has the highest Recall and AP@50–95. Adding modules together does not automatically improve the standalone result.

5.2.2 Four-Class Harbor Architecture Comparison (Condition A)

The principal architecture comparison uses the corrected and frozen Condition A evaluation reference. Table 11 reports all eight configurations. The class columns report AP@50, and all values are percentages.

EfficientViM Block records the highest Condition A mAP@50 and the highest AP for other vessels, whereas ASFF records the highest mAP@50–95 and the baseline records the highest mAP@75. DySample and DySample–ASFF tie for the highest naval and merchant AP, whereas

Table 11: Condition A harbor-four architecture results on the corrected and frozen evaluation reference.

Model	Naval	Merchant	Dock	Other vessel	mAP@50	mAP@75	mAP@50–95
YOLO11n	94.5	76.6	91.7	41.9	76.2	69.8	61.6
+ EfficientViM Block	94.7	76.5	91.7	43.4	76.6	69.6	61.9
+ DySample	95.0	77.0	90.4	42.1	76.1	69.2	61.5
+ ASFF	94.2	76.1	93.0	40.3	75.9	68.8	62.0
+ EfficientViM Block + DySample	94.2	76.2	91.1	40.4	75.5	68.4	60.6
+ EfficientViM Block + ASFF	94.9	76.9	91.5	39.6	75.7	68.7	61.2
+ DySample + ASFF	95.0	77.0	92.0	38.7	75.7	68.9	61.5
+ EfficientViM Block + DySample + ASFF	94.9	76.5	92.5	39.1	75.7	69.0	61.3

ASFF has the highest dock AP. The ordering also differs from the single-class comparison, and none of the configurations improves every metric. The results indicate that no architecture consistently dominates across tasks, classes, and metrics.

The subsequent Condition A/B analysis includes YOLO11n and all three standalone variants because paired annotation-correction results are available for each of them. The later seven-class V0/V1 analysis remains limited to YOLO11n, ASFF, and DySample because no matched EfficientViM Block V0/V1 record is retained. Model inclusion therefore follows the evidence available for each experiment family rather than the ranking in Table 11.

Appendix A.4.2 provides the complete Old GT eight-model table. Because its evaluation labels were later found to be incomplete or inconsistent, it is not used as a parallel leaderboard. The available files also do not establish whether the Old GT and Condition A evaluations used identical frozen checkpoints. The two records show the historical pre-audit and corrected evaluation states but do not provide a causal comparison.

5.3 Annotation Audit, Correction, and Retraining

After the architecture comparison, detector-error inspection exposed a separate source of uncertainty: some visible vessels were missing from the annotations or were labeled inconsistently. This finding motivated the subsequent data-quality analysis.

5.3.1 Annotation Audit and Motivation

Old GT combines the original vessel training annotations with the original validation and test annotations and represents the historical pre-audit state. Its counts of matched detections, background false positives, background false negatives, cross-class confusion, and AP differed from those obtained after the evaluation labels were corrected. The corrected validation and

test vessel labels were frozen before the formal training-label comparison. This fixed reference defines Condition A and is shared unchanged by Condition B.

Identical frozen checkpoints cannot be confirmed for Old GT and Condition A, so their difference is not interpreted as the causal effect of changing evaluation labels. Architecture comparisons instead use the corrected reference.

5.3.2 Human-Assisted Vessel Annotation Correction

The supplied-box vessel-type assistant supports the correction workflow after a human has identified a missing or questionable vessel. It ranks `naval`, `merchant`, and `other_vessel` candidates for the supplied box. It does not localize the vessel, write ground truth, or replace the final annotation decision. Every proposed class requires human confirmation.

A deterministic comparison of the retained training-label snapshots identified 39 matched image-label files whose canonicalized annotation content changed. Across all 2,198 matched files, the vessel-box count changed from 10,860 to 10,936, a net increase of 76. The corresponding net class-count changes were +39 naval, +21 merchant, +3 dock, and +13 other vessel. These are snapshot differences, not counts of individual correction actions. The retained files do not support an unambiguous decomposition into added, deleted, reassigned, and box-adjusted instances. Condition B includes only these vessel-label changes; harbor vehicle annotations belong to the separate seven-class research line.

The independent VC2 evaluation measures this assistance role. Across all 1,571 trusted vessels in Track A, the assistant produced the correct class for 791 objects (0.5035), whereas full-image YOLO produced a correct end-to-end output for 534 objects (0.3399). The difference was 16.36 percentage points (scene-group bootstrap 95% CI: 11.36–21.18 percentage points). As shown in Figure 11, the YOLO panel excludes localization failures from its three-class cells and reports them separately, whereas the assistant panel classifies a supplied reference box for every object. The comparison therefore measures operational annotation-assistance performance under asymmetric inputs rather than general model superiority.

As a secondary ranking result, among the 764 objects missed by YOLO, the assistant placed the correct class first for 342 objects (0.4476) and within the top two for 592 objects (0.7749). These results support human-reviewed candidate ranking, not automatic label acceptance. Detailed calibration, ECE, NLL, Brier scores, frozen temperatures, and confirmation artifacts remain in Appendix A.6.

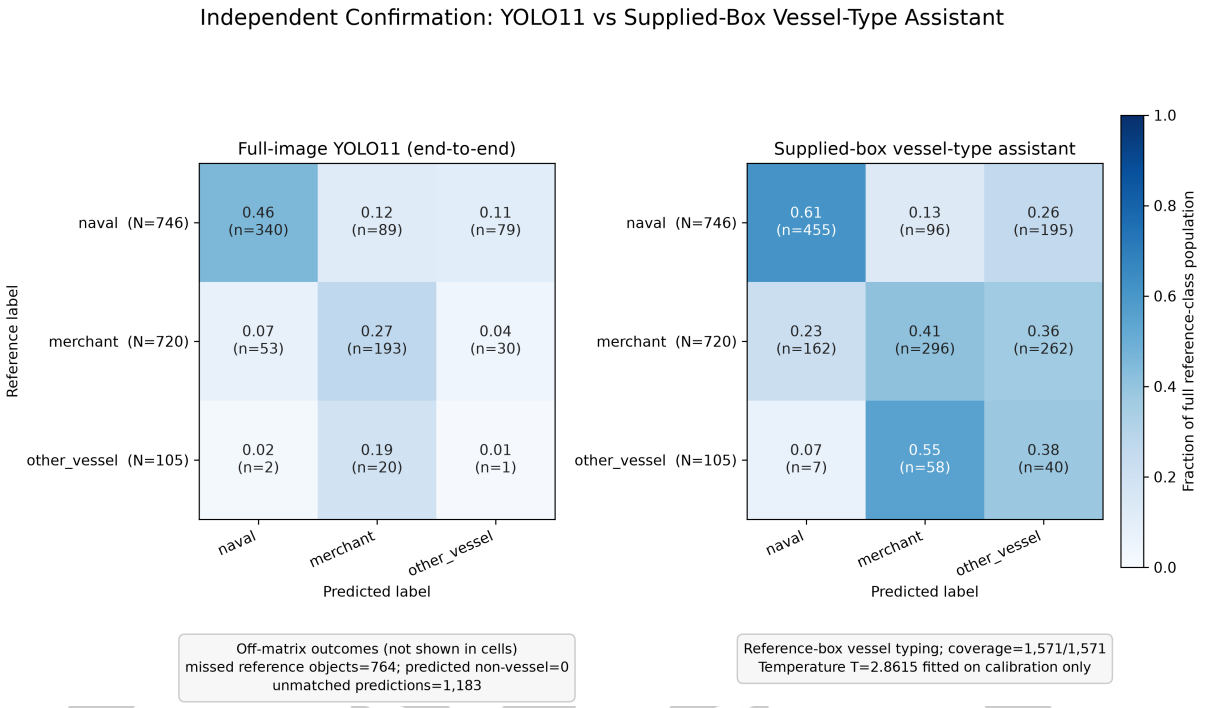


Figure 11: Confusion matrices for the independent VC2 confirmation set. Rows denote reference classes, columns denote predicted classes, and each cell is normalized by the full reference-class population. Localization failures have no predicted vessel class and therefore appear below, rather than inside, the YOLO matrix. The assistant receives a human-supplied reference box and has full classification coverage; the panels consequently represent an operational annotation-assistance comparison, not a same-input test of general model superiority.

Table 12: Primary Track A and secondary Track C results used to evaluate the annotation-assistance role.

Track	System / outcome	N	Correct-class rate
A: all trusted vessels	Full-image YOLO	1,571	0.3399
A: all trusted vessels	Supplied-box assistant	1,571	0.5035
C: YOLO misses	Assistant top-1	764	0.4476
C: YOLO misses	Assistant top-2	764	0.7749

5.3.3 Effect of Training-Label Correction (Condition A vs. Condition B)

Condition A and Condition B use identical corrected and frozen validation and test labels. They differ only in the vessel annotation state of the training data: Condition A uses the original vessel training labels, whereas Condition B uses the partially corrected, human-confirmed vessel training labels. This is the formal comparison for whether training-label correction changed detector behavior.

Table 13: Condition A and Condition B results. Class columns report AP@50; all values are percentages.

Model	Condition	Naval	Merchant	Dock	Other vessel	mAP@50	mAP@75	mAP@50–95
YOLO11n	A	94.5	76.6	91.7	41.9	76.2	69.8	61.6
YOLO11n	B	94.3	76.9	92.1	45.0	77.0	69.3	62.2
EfficientViM Block	A	94.7	76.5	91.7	43.4	76.6	69.6	61.9
EfficientViM Block	B	94.9	77.6	91.6	44.0	77.1	69.8	62.0
DySample	A	95.0	77.0	90.4	42.1	76.1	69.2	61.5
DySample	B	94.7	77.9	91.4	44.8	77.2	70.1	62.6
ASFF	A	94.2	76.1	93.0	40.3	75.9	68.8	62.0
ASFF	B	94.7	77.0	91.6	42.1	76.3	69.0	61.6

Table 14: Condition A to Condition B AP changes in percentage points.

Model	Naval	Merchant	Dock	Other vessel	ΔmAP@50	ΔmAP@75	ΔmAP@50–95
YOLO11n	-0.2	+0.3	+0.4	+3.1	+0.8	-0.5	+0.6
EfficientViM Block	+0.2	+1.1	-0.1	+0.6	+0.5	+0.2	+0.1
DySample	-0.3	+0.9	+1.0	+2.7	+1.1	+0.9	+1.1
ASFF	+0.5	+0.9	-1.4	+1.8	+0.4	+0.2	-0.4

The response is architecture-dependent. YOLO11n increases by 3.1 points for other vessels, 0.8 points in mAP@50, and 0.6 points in mAP@50–95, while its mAP@75 decreases by 0.5 points. EfficientViM Block increases by 0.5 points in mAP@50, 0.2 points in mAP@75, and 0.1 points in mAP@50–95; its largest class-level change is a 1.1-point increase for merchant vessels, while dock AP decreases by 0.1 points. DySample improves after correction: AP for merchant vessels, docks, and other vessels increases by 0.9, 1.0, and 2.7 points, respectively; mAP@50 and mAP@50–95 each increase by 1.1 points, and mAP@75 increases by 0.9 points. Its naval AP decreases by 0.3 points. ASFF increases by 0.4 points in mAP@50 and 0.2 points in mAP@75 but decreases by 0.4 points in mAP@50–95. Thus, all four architectures improve in mAP@50, but the effect is not uniform across every class, aggregate metric, and operating-point error type.

Table 15 juxtaposes the retained matrix raw-count changes with the corrected summary-image AP changes. Both artifact types use the same condition definitions and corrected evaluation reference, but the copied files do not preserve enough metadata to prove a common validation-directory identity for every model–condition row. The matrix convention is fixed: columns are true classes and rows are predicted classes. The background column counts predictions on true background as false positives, and the background row counts true objects predicted as background as false negatives.

Table 15: Condition A to Condition B fixed-artifact error-count changes and separately reported mAP changes. Background FP and FN are distinct from cross-class confusion.

Model	ΔTP	ΔBackground FP	ΔBackground FN	ΔCross-class confusion	ΔmAP@50	ΔmAP@75	ΔmAP@50–95
YOLO11n	+56	+99	-48	-8	+0.8	-0.5	+0.6
EfficientViM Block	+17	+49	-17	0	+0.5	+0.2	+0.1
ASFF	0	-7	-24	+24	+0.4	+0.2	-0.4
DySample	+20	-44	-1	-19	+1.1	+0.9	+1.1

For YOLO11n, total true positives increased from 2,924 to 2,980 and background false negatives decreased from 483 to 435, but background false positives increased from 576 to 675, while cross-class confusion decreased from 268 to 260. Thus, partial training-label correction improved detection coverage and reduced missed detections, but the gain was accompanied by more predictions on true background. Correction did not reduce every error type.

For EfficientViM Block, diagonal true positives increased from 2,923 to 2,940 and background false negatives decreased from 485 to 468. Background false positives, however, increased from 602 to 651, while cross-class confusion remained unchanged at 267. These operating-point counts are consistent with greater object coverage accompanied by more predictions on true background; the three aggregate AP measures nevertheless increased slightly.

For DySample, mAP@50 and mAP@50–95 increased by 1.1 points, while mAP@75 increased by 0.9 points. At the retained matrix operating point, true positives increased from 2,896 to 2,916, background false positives decreased from 683 to 639, background false negatives changed only from 492 to 491, and cross-class confusion decreased from 287 to 268. The aggregate operating-point counts move in the same favorable direction as AP, although the class-level response remains mixed.

For ASFF, diagonal true positives remained at 2,909, while background false positives decreased from 609 to 602 and background false negatives decreased from 484 to 460. At the same time, off-diagonal class confusions increased from 282 to 306; mAP@50 increased by 0.4 points and mAP@75 by 0.2 points, whereas mAP@50–95 decreased by 0.4 points. The

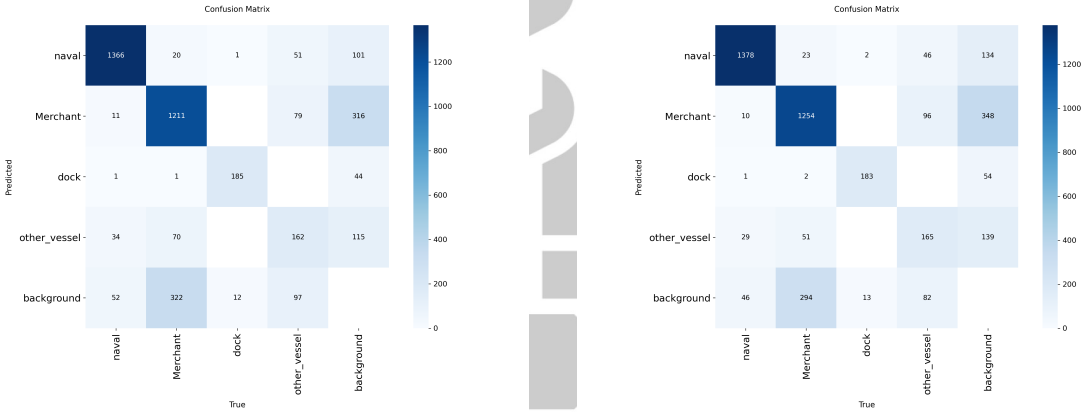
ASFF result represents a redistribution of errors rather than a uniform performance gain. This is plausible because AP integrates ranked detections and localization quality, whereas the matrix summarizes one operating point.

The YOLO11n class-level evidence most directly related to vessel correction is shown in Table 16. The merchant class gained 43 true positives and 28 fewer background misses, but also produced 32 additional background false positives. The class for other vessels gained three true positives and 15 fewer background misses, with 24 additional background false positives.

Table 16: YOLO11n background-error counts for the two vessel classes most affected by correction.

Class	Condition	TP	Background FP	Background FN	AP@50
Merchant	A	1,211	316	322	76.6
Merchant	B	1,254	348	294	76.9
Other vessel	A	162	115	97	41.9
Other vessel	B	165	139	82	45.0

Figure 12 shows the corresponding YOLO11n matrices under the same fixed corrected evaluation reference.



(a) Condition A. (b) Condition B.

Figure 12: Confusion matrices of YOLO11n under Conditions A and B using the fixed corrected evaluation reference. Columns are true classes and rows are predicted classes; values are raw counts.

The corresponding EfficientViM Block, ASFF, and DySample pairs are provided in Figures 16, 17, and 18 in the appendix.

5.4 Targeted Harbor-Domain Supervision in the Corrected Seven-Class Setting

5.4.1 Combined Harbor-Domain Treatment Design (V0 vs. V1)

The corrected harbor vessel-label state and the airport-derived data define the corrected seven-class harbor–airport setting in Table 8. V0 is the baseline in this setting. It contains no added harbor vehicle supervision and no shipboard-aircraft foreground-composited training samples. V1 starts from the same setting and simultaneously adds three forms of supervision: manual harbor-vehicle annotations, foreground-composited harbor vehicles, and foreground-composited shipboard aircraft.

The V0-to-V1 comparison evaluates the complete harbor-domain treatment. Because the design contains no factorial or component-wise ablation, an observed change cannot be attributed separately to manual vehicle annotation, harbor-vehicle compositing, or shipboard-aircraft compositing. The historical labels C0 and C1 refer to a different airplane-only experiment documented in Appendix A.4.1.

The final V0/V1 results are recorded in the six evaluation outputs `val107`–`val112`. All report the same 1,087-image, 11,374-instance test population. The corresponding training-run directories, dataset YAML files, and checkpoints are unavailable, so the numerical tables below reproduce these evaluation outputs rather than a reconstructed run.

5.4.2 Harbor-Vehicle and Shipboard-Aircraft Supervision (V1)

Table 17 separates the types of supervision present in V1 without treating them as independent experimental conditions. Manual annotation contributes 48 boxes labeled as large vehicle and 660 labeled as small vehicle. Harbor-vehicle foreground compositing adds 53 boxes labeled as large vehicle and 15 labeled as small vehicle. Together these sources add 101 large vehicles and 675 small vehicles, or 776 harbor-vehicle boxes, across a verified combined membership of 26 affected harbor training images. Shipboard-aircraft foreground compositing additionally contributes 61 airplane boxes across 34 training host images.

Separate unique-image counts are unavailable for the manual and foreground-composited harbor-vehicle sources; only their combined 26-image membership is known. These descriptions show how the V1 data were constructed but do not define separate experimental contrasts.

Table 17: Supervision components present simultaneously in the V1 training dataset.

Added supervision type	Host images	Airplane	Large vehicle	Small vehicle	Vehicle total
Manual harbor-vehicle annotation	–	–	48	660	708
Harbor-vehicle foreground compositing	–	–	53	15	68
Combined harbor-vehicle addition	26	–	101	675	776
Shipboard-aircraft foreground compositing	34	61	–	–	–

5.4.3 Global Seven-Class Evaluation

Table 18 reports the final global V0/V1 mixed-domain results. Class columns show AP@50 and all values are percentages. These values come from the final six evaluation outputs, not from the historical C0/C1 experiment.

Table 18: Seven-class detection performance under V0 and the combined V1 treatment; all AP and mAP values are percentages.

Model	Cond.	Naval	Merchant	Dock	Other	Airplane	Large veh.	Small veh.	mAP@50	mAP@50–95
YOLO11n	V0	93.7	76.3	91.9	37.0	75.4	54.5	66.7	70.8	50.4
+ ASFF	V0	94.3	74.7	89.4	38.3	72.1	55.4	66.7	70.1	50.2
+ DySample	V0	93.5	76.8	90.0	36.5	74.5	51.9	65.9	69.9	49.9
YOLO11n	V1	94.1	76.6	90.4	39.0	75.1	57.3	67.2	71.4	51.1
+ ASFF	V1	94.5	77.5	90.2	39.6	72.6	56.1	67.7	71.2	50.8
+ DySample	V1	94.0	77.7	85.7	38.8	74.3	54.5	66.1	70.2	50.1

Across the paired V0–V1 results, both aggregate metrics and both vehicle-class AP values increase for all three architectures. Table 19 summarizes these changes. YOLO11n increases from 70.8 to 71.4 in mAP@50 and from 50.4 to 51.1 in mAP@50–95. ASFF increases from 70.1 to 71.2 and from 50.2 to 50.8, while DySample increases from 69.9 to 70.2 and from 49.9 to 50.1. AP@50 for both large and small vehicles also increases under V1 for every architecture.

Table 19: Changes in vehicle-class AP and aggregate mAP from V0 to the combined V1 treatment, in percentage points.

Model	Δ Large vehicle	Δ Small vehicle	ΔmAP@50	ΔmAP@50–95
YOLO11n	$+2.8$	$+0.5$	$+0.6$	$+0.7$
ASFF	$+0.7$	$+1.0$	$+1.1$	$+0.6$
DySample	$+2.6$	$+0.2$	$+0.3$	$+0.2$

The aggregate changes are positive but small, and not every class improves. For example, YOLO11n AP@50 for dock and airplane decreases by 1.5 and 0.3 points, respectively, and DySample AP@50 for those classes decreases by 4.3 and 0.2 points. The result shows a consistent descriptive response to the combined V1 package in aggregate mAP and vehicle-class AP,

with mixed changes among individual non-target classes. It does not show which V1 component produced any particular change.

5.4.4 Harbor-Specific Vehicle Evaluation

The harbor-specific evaluation consists of 12 images and 383 instances, including 17 large vehicles and 298 small vehicles, as reported consistently by all six outputs. Its dataset YAML, directory, and file-membership manifest are unavailable, so this small evaluation is treated as a pilot record rather than reconstructed from source files. An earlier 6-image validation count cannot be confirmed and is not used in the analysis.

Table 20 reports the vehicle response on this 12-image record. The overall columns reproduce the aggregate values for the complete seven-class subset; they are not an average of only the two vehicle rows.

Table 20: Harbor-specific vehicle response under V0 and the combined V1 treatment; all metric values use the 0–1 scale.

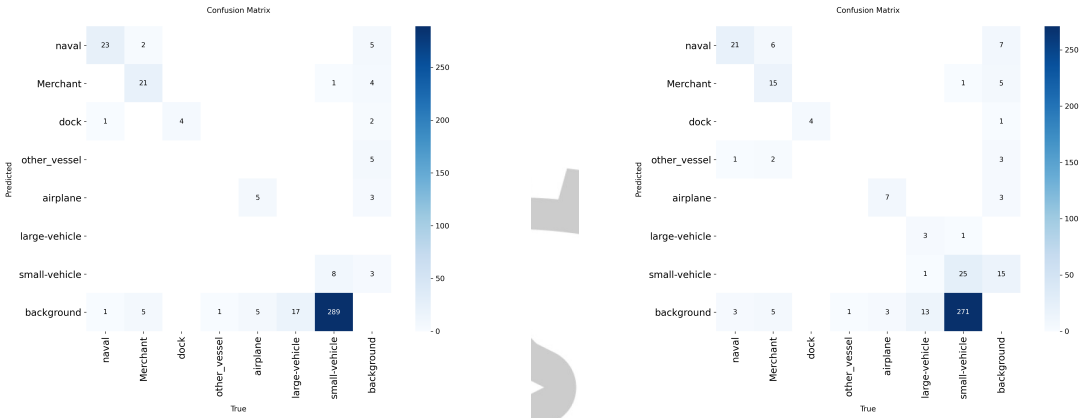
		Large vehicle			Small vehicle			Overall subset	
Model	Cond.	Recall	AP@50	AP@50–95	Recall	AP@50	AP@50–95	mAP@50	mAP@50–95
YOLO11n	V0	0.0000	0.0000	0.0000	0.0201	0.288	0.1130	0.490	0.365
YOLO11n	V1	0.1760	0.1540	0.0818	0.0631	0.322	0.1210	0.532	0.376
+ ASFF	V0	0.0000	0.0156	0.0140	0.0118	0.277	0.1040	0.460	0.321
+ ASFF	V1	0.1760	0.1860	0.0915	0.0121	0.271	0.1070	0.510	0.371
+ DySample	V0	0.0000	0.0000	0.0000	0.0114	0.302	0.1210	0.470	0.354
+ DySample	V1	0.1760	0.1980	0.0949	0.0738	0.208	0.0677	0.527	0.377

All three architectures have higher aggregate AP in the harbor-specific evaluation under V1. YOLO11n increases from 0.490 to 0.532 in mAP@50 and from 0.365 to 0.376 in mAP@50–95. ASFF changes from 0.460 to 0.510 and from 0.321 to 0.371, while DySample changes from 0.470 to 0.527 and from 0.354 to 0.377. This targeted pattern complements the smaller positive changes observed on the full mixed-domain test.

Large vehicle detectability changes from nearly absent to measurable for all three architectures: V1 Recall is 0.176 in each case and both AP measures increase. Because the test contains only 17 large vehicle instances, this pattern is descriptive and does not support statistical or generalization claims. Small vehicle behavior is mixed. YOLO11n improves Recall, AP@50, and AP@50–95; ASFF remains nearly stable; and DySample increases Recall from 0.0114 to 0.0738 while AP@50 decreases from 0.302 to 0.208 and AP@50–95 decreases from 0.121 to 0.0677. Thus, DySample becomes more sensitive at the reported operating point but has poorer ranked precision–recall quality. All of these are vehicle responses under the combined V1 treat-

ment; they cannot be assigned solely to the vehicle-supervision components because V1 also contains shipboard-aircraft training additions.

Figure 13 compares the corresponding YOLO11n confusion matrices. Their ground-truth counts and evaluation sequence identify the 12-image test, and the V0/V1 filename tokens agree with the condition definitions.

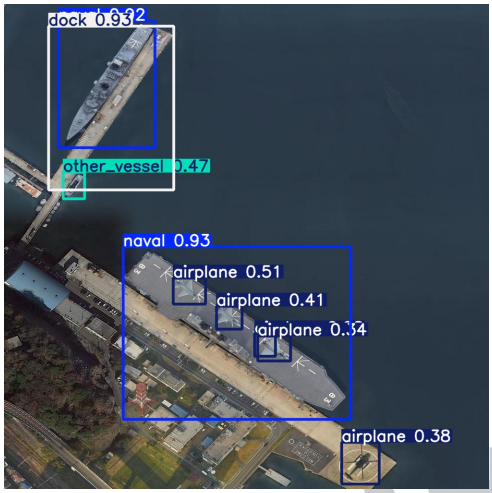


(a) V0. (b) V1.

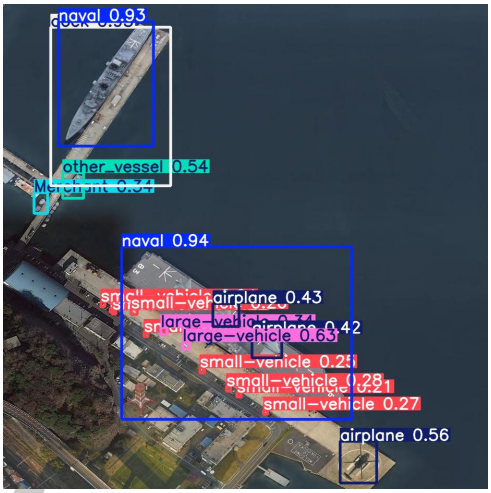
Figure 13: YOLO11n confusion matrices under the corrected baseline (V0) and combined treatment (V1) on the 12-image harbor-specific test.

The full seven-class and harbor-specific evaluations provide complementary views. On the global mixed-domain test, V1 produces consistent positive changes in aggregate mAP and in both vehicle-class AP values across YOLO11n, ASFF, and DySample. The targeted harbor evaluation further shows that large vehicle detection changes from nearly absent to measurable across all three architectures. Small vehicle behavior is more architecture-dependent: YOLO11n improves in Recall and both AP measures, ASFF remains approximately stable, and DySample exhibits a Recall–AP trade-off.

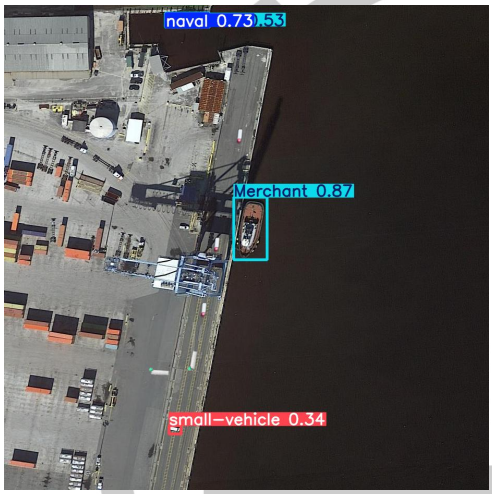
Figure 14 compares YOLO11n predictions on three matched harbor scenes. Relative to V0, V1 produces additional large- and small-vehicle predictions on carrier decks, quays, and adjacent road areas. The scenes illustrate the detector's response to the combined treatment but do not constitute a representative accuracy estimate. Because no ground-truth overlay is shown, the individual added predictions cannot be assessed as correct or incorrect from these images alone.



(a) Scene 1, V0.



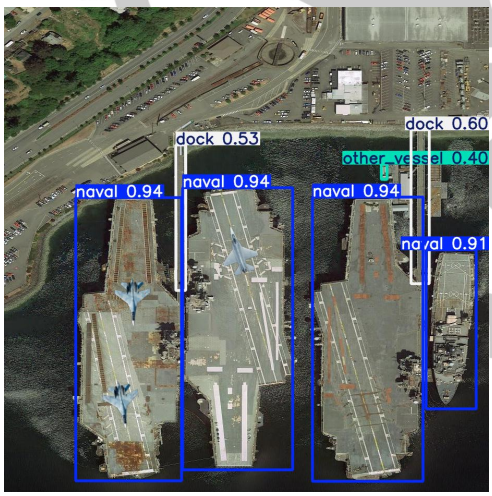
(b) Scene 1, V1.



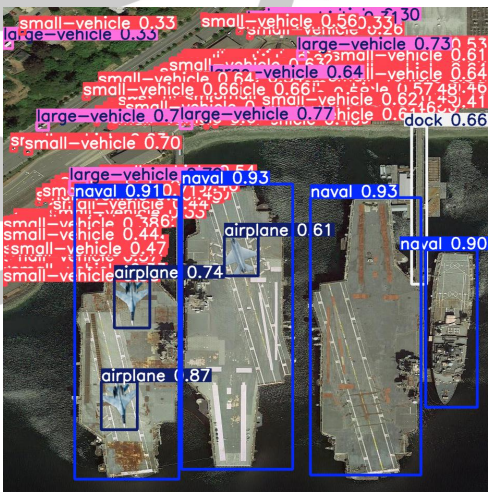
(c) Scene 2, V0.



(d) Scene 2, V1.



(e) Scene 3, V0.



(f) Scene 3, V1.

Figure 14: YOLO11n predictions on three matched harbor scenes under the corrected baseline (V0) and combined treatment (V1). V1 produces additional large- and small-vehicle predictions on decks, quays, and adjacent road areas. These examples are qualitative and do not provide an independent estimate of accuracy.

5.4.5 Shipboard-Aircraft Evaluation

The harbor source contains no `airplane` supervision and no airplane–vessel relationship. In the final design, the 34 training hosts and 61 composited airplane boxes are part of V1 itself, together with both types of harbor-vehicle supervision. For consistency, the airplane response is evaluated using the same V0 and V1 conditions as the global comparison.

Table 21: Airplane response on the final standard seven-class evaluation under V0 and the combined V1 treatment; all metric values use the 0–1 scale.

Model	Condition	Recall	AP@50	AP@50–95
YOLO11n	V0	0.632	0.754	0.542
YOLO11n	V1	0.627	0.751	0.542
ASFF	V0	0.636	0.721	0.527
ASFF	V1	0.636	0.726	0.522
DySample	V0	0.619	0.745	0.529
DySample	V1	0.639	0.743	0.534

The standard-test airplane rows are transcribed from the same six final evaluation outputs used in Table 18. The response is mixed: YOLO11n AP@50 decreases by 0.003 and AP@50–95 is unchanged; ASFF AP@50 increases by 0.005 while AP@50–95 decreases by 0.005; and DySample Recall and AP@50–95 increase while AP@50 decreases by 0.002. These values describe airplane behavior under the combined V1 package and do not estimate the isolated effect of aircraft compositing.

Table 22: Constructed shipboard-aircraft test-set identity and final-result availability.

Evaluation	Images	Airplanes	Final V0/V1 result status
Fixed constructed shipboard test	11	25	Dataset membership is verified, but no evaluation artifact tied to the final V0/V1 checkpoints is present in the supplied archive.

The preserved files show that the constructed test contains 11 images and 25 airplane instances. The final V0/V1 checkpoints are unavailable, however, and the existing predictions were generated by an older, airplane-only C0/C1 pair. Appendix A.4.1 preserves those historical results, but they are not substituted for the final experiment. The response on the constructed shipboard test under the combined V0/V1 treatment remains unquantified.

5.4.6 Integrated Interpretation

The final V0/V1 design evaluates a single data-side package intended to address two harbor-domain deficits: limited vehicle supervision for objects already present in harbor imagery and the absence of a shipboard-aircraft relationship. Across the three architectures, the global evaluation shows small positive aggregate changes and higher AP@50 for both vehicle classes, although results for individual non-vehicle classes, including airplane, are mixed. The small harbor-specific test also shows a measurable large vehicle response, but its membership manifest is unavailable. The constructed shipboard set has been recounted, but no result matched to the final V0/V1 checkpoints is available.

The paired comparison supports a treatment-level interpretation only. It cannot identify the contribution of an individual component, and the legacy C0/C1 shipboard results cannot substitute for the missing final evaluation.

5.5 Integrated Findings and Study Limitations

The architecture and annotation results form a sequential account. The module comparisons showed task-, class-, and metric-dependent trade-offs rather than a universal architecture. Error inspection then showed that annotation quality also affected detector behavior. Under a shared corrected evaluation reference, partial vessel-label correction improved mAP@50 for all four retained A/B architectures. YOLO11n improved most clearly for `other_vessel` and missed detections, although it also increased background false positives and its mAP@75 decreased. EfficientViM Block produced small gains in all three aggregate AP measures and more true positives, but also more background false positives. DySample improved by 1.1 points in both mAP@50 and mAP@50–95 while its aggregate operating-point counts also improved. ASFF was mixed: background errors decreased and mAP@50 and mAP@75 increased slightly, but class confusions increased and mAP@50–95 decreased. Data quality affects detector behavior, but the size and form of the change still depend on architecture, class, metric, and operating point.

The combined V0/V1 experiment addresses insufficient vehicle supervision and an absent shipboard-aircraft co-occurrence within one V1 training dataset. On the full mixed-domain seven-class test, all three architectures show higher aggregate mAP and higher AP for both large and small vehicles under V1. Individual non-vehicle classes remain mixed, including the

airplane response, so this is a descriptive treatment-level pattern rather than an improvement in every class. On the harbor-specific test, all three architectures show higher aggregate AP, and large vehicle detection changes from nearly absent to measurable. Small vehicle behavior is architecture-dependent: YOLO11n improves in Recall and AP, ASFF remains approximately stable, and DySample increases Recall while AP decreases. These vehicle results describe responses under the complete V1 package, not the isolated effect of vehicle annotation or vehicle compositing.

The constructed shipboard test contains 11 images and 25 airplane instances. Its preserved predictions, however, were generated by the older airplane-only C0/C1 checkpoints rather than the final V0/V1 checkpoints. Appendix A.4.1 documents those historical values. The final standard-test airplane response is mixed, while the constructed-shipboard response under the combined V0/V1 treatment remains unavailable.

Several limitations constrain these findings. Vessel training correction is partial: 39 of 2,198 matched training-label files changed, and the total box count increased from 10,860 to 10,936. The available snapshots do not permit an unambiguous action-level decomposition of every change into added, deleted, reassigned, and adjusted boxes. Each formal detector condition is represented by one seed, so the reported changes are descriptive. The harbor-specific vehicle test contains only 12 images, including 17 large vehicle instances, so its consistent descriptive increase should be treated as a pilot result rather than evidence of statistical significance or broad generalization. Its dataset YAML and file-membership manifest are unavailable. V1 contains manual harbor-vehicle annotation, harbor-vehicle foreground compositing, and shipboard-aircraft foreground compositing simultaneously. Because no factorial or component-wise ablation was completed, their individual contributions cannot be separated. The paired experiment can be interpreted only at the treatment level. The constructed shipboard test has 11 images and 25 airplane instances, the final checkpoint-matched evaluation is absent, no independent real shipboard-aircraft test is available, and real-scene transfer remains untested. Finally, the observed responses differ by architecture, class, metric, evaluation set, and operating point and must not be generalized to every detector.

Chapter 6. Conclusion and Future Work

6.1 Research Summary

This thesis investigated cross-domain object detection using harbor and airport-derived data. The study compared three YOLO11n architecture interventions, examined detector errors and vessel-annotation quality, evaluated partial correction of vessel training labels under a fixed reference, and then tested a combined harbor-domain supervision treatment in a corrected seven-class setting. The treatment added manual harbor-vehicle annotations, foreground-composited harbor vehicles, and foreground-composited shipboard aircraft to the same training dataset. GUI-based tools supported data construction, annotation review, inference, visualization, analysis, and reporting.

The architecture results were task-dependent, and annotation correction affected both average precision and error behavior. The combined treatment produced positive aggregate and harbor-vehicle responses, particularly for large vehicles, while small vehicle and standard-test airplane results varied by architecture. The constructed shipboard test set was verified, but the final experiment checkpoints were unavailable for evaluating it.

6.2 Conclusions by Research Question

The following conclusions answer the four research questions introduced in Section 1.4. Because most detector conditions were evaluated with a single run, the comparisons are descriptive.

RQ1 – Architecture behavior.

The architecture interventions produced trade-offs that depended on the task, class, metric, and computational cost; no YOLO11n variant was consistently superior. Under Condition A, EfficientViM Block achieved the highest mAP@50 at 76.6, the baseline achieved the highest

mAP@75 at 69.8, and ASFF achieved the highest mAP@50–95 at 62.0. Class-level leaders also differed, and module combinations did not yield uniformly additive gains. Architecture selection should therefore reflect the target task and acceptable cost rather than a single table entry. These single-run comparisons do not establish statistical superiority.

RQ2 – Annotation quality and correction.

Partial correction of vessel training labels changed both detector performance and error behavior under the fixed evaluation reference. YOLO11n mAP@50 increased from 76.2 to 77.0, mAP@50–95 from 61.6 to 62.2, and AP@50 for other vessels from 41.9 to 45.0, although mAP@75 decreased from 69.8 to 69.3. DySample gained 1.1 points in both mAP@50 and mAP@50–95, while ASFF gained 0.4 points in mAP@50 but lost 0.4 points in mAP@50–95.

EfficientViM Block also showed a small positive aggregate response: mAP@50 increased from 76.6 to 77.1, mAP@75 from 69.6 to 69.8, and mAP@50–95 from 61.9 to 62.0. At class level, merchant AP increased by 1.1 points, whereas dock AP decreased by 0.1 points.

The confusion matrices show that correction redistributed errors differently across architectures. For YOLO11n, true positives increased by 56 and background false negatives decreased by 48, but background false positives increased by 99. EfficientViM Block gained 17 true positives and reduced background false negatives by 17, while adding 49 background false positives and leaving cross-class confusion unchanged. DySample showed a more consistently favorable aggregate shift, whereas ASFF reduced background errors but increased cross-class confusion. Correction therefore did not improve every metric or error type. The Old GT comparison motivates the corrected reference but is not part of the training-label treatment.

RQ3 – Combined harbor-domain supervision.

The combined V1 treatment produced small positive aggregate changes and measurable harbor-vehicle responses. On the global seven-class test, mAP@50 increased by 0.6 points for YOLO11n, 1.1 for ASFF, and 0.3 for DySample; mAP@50–95 and AP@50 for both vehicle classes also increased for all three architectures. Results for individual non-vehicle classes did not all improve.

The harbor-specific test contains 12 images, 17 large vehicles, and 298 small vehicles. Recall for large vehicles increased from 0 to 0.176 for every architecture, and AP@50 rose to 0.154 for YOLO11n, 0.186 for ASFF, and 0.198 for DySample. Small vehicle results were architecture-dependent: YOLO11n improved, ASFF remained approximately stable, and DySample gained Recall while losing AP. The standard-test airplane response was also mixed.

These results describe the complete V1 treatment, which simultaneously includes manual harbor-vehicle annotation, harbor-vehicle foreground compositing, and shipboard-aircraft foreground compositing. Without a component-wise ablation, the effect of any one component cannot be isolated. The constructed shipboard test contains 11 images and 25 airplanes, but no result tied to the final V0/V1 checkpoints is available. The older C0/C1 results come from an airplane-only intervention and a different standard test, so they remain historical rather than final evidence. The small harbor-specific test, unavailable membership manifest, missing final constructed-test result, and absence of an independent real shipboard test limit broader interpretation.

RQ4 – Research-support systems.

The GUI-based systems make data construction, annotation review, inference, and result inspection more traceable while preserving human control over semantic decisions. The dataset interface supports annotation inspection, RGBA handling, Smart Zone placement, transformation preview, synchronized labels, and final acceptance. On 1,571 trusted vessels, supplied-box assistant accuracy was 0.5035 compared with 0.3399 for full-image YOLO, a descriptive difference of 16.36 percentage points in the operational comparison. As a secondary ranking measure, among 764 vessels missed by YOLO, assistant top-1 and top-2 accuracy were 0.4476 and 0.7749. These results support its use as a human-confirmed decision aid for missing-label review rather than as an autonomous replacement for YOLO. The post-training platform links registered-model inference to visualization, history, auxiliary analysis, and Markdown/PDF export.

The assistant ranks classes only after the researcher supplies a box; it does not localize vessels, write ground truth, or replace human judgement. The software evaluations address research workflow functions, not production readiness, real-time throughput, detector correctness, or report correctness.

6.3 Research Contributions

The completed work makes four contributions aligned with Chapter 1.

1. **Cross-domain seven-class data setting.** Four harbor and three airport-derived classes are integrated while preserving their source domains. Direct source-data recounts and documented evaluation records enable global and domain-specific analysis without pre-

senting the setting as a new public benchmark.

2. **Data-centric correction and construction workflow.** Vessel-label correction, added vehicle supervision, and shipboard-aircraft foreground compositing use synchronized annotations and human review. The thesis distinguishes annotation quality, insufficient class/domain supervision for existing objects, and an absent target-context representation; foreground compositing is not claimed as novel.

3. **Comparative experiments.** The experiments show that no architecture wins universally, quantify the effects of partial vessel training-label correction, and report global, harbor-vehicle, and standard-test airplane responses under the combined V0/V1 treatment. The historical C0/C1 experiment remains separate from the final results, and no state-of-the-art claim is made.

4. **GUI-based research support.** A dataset and annotation system, human-confirmed vessel assistant, and post-training platform support traceable construction and analysis while leaving semantic decisions and report review to the researcher.

6.4 Research Limitations

Experimental repetition and statistical strength. Most detector comparisons use one run and are descriptive. The historical three-seed baseline applies only to the earlier harbor-four setting, not the newer conditions.

Partial annotation correction. The retained correction covers 39 of 2,198 matched training-label files, with the vessel-box count changing from 10,860 to 10,936. It remains a partial intervention, and the snapshots do not permit an unambiguous action-level decomposition of every annotation change. Condition A/B therefore does not represent a complete audit of the vessel dataset.

Scale, provenance, and causal attribution. The harbor-specific test has 12 images, 17 large vehicles, and 298 small vehicles; it is a pilot evaluation, and its dataset YAML and membership manifest are unavailable. V1 contains multiple data-side interventions simultaneously. Because no factorial or component-wise ablation was completed, the observed V0/V1 changes cannot be attributed separately to manual harbor-vehicle annotation, harbor-vehicle foreground compositing, or shipboard-aircraft foreground compositing. The paired comparison can therefore be interpreted only at the level of the complete treatment.

Constructed shipboard domain. The test has 11 constructed images and 25 airplanes, but the final V0/V1 checkpoints and their matched constructed-test result are not retained. No real counterpart is available, so synthetic-to-real generalization is untested. System evidence covers workflow functions, not operator efficiency, deployment readiness, or report correctness.

6.5 Future Work

Complete and repeat annotation correction. The vessel-label audit should be extended beyond the retained 39-image partial intervention, followed by multi-seed Condition A/B runs under the same reference to test whether the observed patterns persist.

Expand and separate targeted-supervision evidence. A larger, diverse held-out set should include more large vehicles and broader scene conditions. A factorial or component-wise ablation should separately evaluate manual harbor-vehicle annotation, harbor-vehicle foreground compositing, and shipboard-aircraft foreground compositing.

Complete shipboard-aircraft evidence. The final V0/V1 checkpoints should first be evaluated on the verified constructed set under a recorded command, followed by an independent real evaluation. Future synthetic conditions need source-disjoint hosts and foregrounds, diverse assets, recorded provenance, greater scene realism, and human review.

Improve and evaluate research support. User studies should measure annotation time, accuracy, usability, and failures. The assistant needs broader untouched evaluation, exports should preserve provenance, and reports require human factual review.

These findings suggest that architectural modification need not be the first response to performance limitations in cross-domain object detection. In the evaluated harbor–airport setting, inspection of detector errors revealed incomplete vessel annotations and gaps in task-relevant supervision that model redesign alone could not supply. Partial label correction and the combined targeted-supervision treatment improved selected outcomes, although their effects varied by architecture, class, metric, and evaluation set; the study does not establish that data refinement generally outperforms architecture changes. The human-supervised tools make the associated data construction and analysis more inspectable. Repeated experiments, component-wise ablations, complete artifact preservation, and independent real-domain tests are needed to test and extend these findings.

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Appendix A. Supplementary and Reproducibility Records

A.1 Dataset and Condition Records

The experiments use the canonical seven-class taxonomy in Table 8. Harbor source IDs 0–3 retain their order, whereas airport source IDs 0–2 are offset by four to joint IDs 4–6. The resulting class names and source groups are listed in the referenced table. The condition names identify controlled comparisons rather than a cumulative experimental sequence. Condition A/B concerns vessel training-label quality. The corrected harbor vessel-label state is combined with the airport data to define V0, and V1 adds the complete harbor-domain supervision package to that baseline.

Table 23: Condition identities used by the formal comparisons.

Comparison	Condition pair	Controlled difference
Annotation correction	Condition A / Condition B	Original versus partially corrected vessel training labels; both use the same corrected and frozen evaluation reference.
Combined targeted supervision	V0 / V1	V0 has the corrected seven-class data and no added harbor-domain supervision. V1 simultaneously adds manual harbor-vehicle annotations, foreground-composited harbor vehicles, and foreground-composited shipboard aircraft.

The formal comparisons are Condition A versus Condition B and V0 versus V1. C0/C1 and B0/D0 identify a historical airplane-context experiment and do not define additional scientific conditions.

Table 9 defines the role of each condition. The retained evaluation identities are listed in Table 24. The table distinguishes artifact-backed records from file-recounted datasets.

The final V0/V1 evaluation screenshots preserve the numerical results but not the corresponding immutable training datasets and checkpoints. By contrast, the historical C0/C1 dataset

Table 24: Retained evaluation identities for the final V0/V1 experiment.

Record	Population	Provenance status
Final standard V0/V1	1,087 images; 11,374 instances; 938 airplanes; 1,651 large vehicles; 5,174 small vehicles	Six consistent evaluation outputs; final dataset YAML and checkpoint files are not retained.
Harbor-specific V0/V1	12 images; 383 instances; 17 large vehicles; 298 small vehicles	Six consistent evaluation outputs; dataset YAML and membership manifest are not retained.
Constructed shipboard set	11 images; 25 airplanes	Dataset membership and labels verified from preserved files; final V0/V1 checkpoint-matched metrics are unavailable.

has a complete builder, YAML, membership manifest, hash registry, and checkpoint pair. Because it implements an airplane-only treatment, it is documented separately in Section A.4.1.

A.2 Annotation-Correction Supplementary Evidence

Condition A uses the original vessel training labels, whereas Condition B uses the partially corrected, human-confirmed vessel training labels. Both conditions use the same corrected and frozen validation and test annotations. The treatment changes the training-label state without changing the evaluation reference or introducing the harbor vehicle and shipboard-aircraft combined targeted supervision studied under V0/V1.

The correction workflow begins when a researcher identifies a missing or questionable vessel annotation. The supplied-box assistant may rank the three vessel-type candidates, but it neither localizes the vessel nor writes ground truth. Only a human-confirmed box and class may enter the corrected training state. The assistant's frozen evaluation is documented in Section A.6.

The original and corrected snapshots each contain 2,198 matched training-label files. A corrected training image is operationally defined as a matched label file whose parsed annotation multiset differs after numeric canonicalization; whitespace, line order, and insignificant numeric formatting are ignored. This deterministic comparison identified 39 changed image-label files. Table 25 reports the direct snapshot counts.

Table 25: Verified original and corrected vessel training-label snapshot counts.

Metric	Original	Corrected	Net change
Matched label files	2,198	2,198	0
Vessel boxes	10,860	10,936	+76
Naval boxes	4,583	4,622	+39
Merchant boxes	4,387	4,408	+21
Dock boxes	744	747	+3
Other vessel boxes	1,146	1,159	+13

These net differences do not identify correction actions. No action log or project-specific matching rule is available, so the snapshots cannot be reliably decomposed into added, deleted, reassigned, and box-adjusted instances. Detailed label-file differences remain in the project archive. The Condition A/B results and error-count changes are reported in Section 5.3.3.

A.3 Combined Harbor-Domain Supervision Records (V1)

V1 is one treatment containing added supervision for harbor vehicles and the airplane–vessel co-occurrence absent from the source harbor data. Table 26 lists the added training data and the corresponding evaluation populations.

Table 26: Added V1 supervision and targeted evaluation populations.

Condition	Record role	Source	Images	Airplane	Large vehicle	Small vehicle
V1	Training supervision	Manual harbor annotation	–	–	48	660
V1	Training supervision	Harbor foreground compositing	–	–	53	15
V1	Combined vehicle total	Both vehicle sources	26	–	101	675
V1	Training supervision	Shipboard-aircraft compositing	34	61	–	–
V0/V1	Retained harbor test	Evaluation artifact	12	10	17	298
V0/V1	Constructed shipboard test	Verified dataset files	11	25	–	–

The available data identify 26 affected harbor training images in total but do not provide separate image counts for the manual and foreground-composited vehicle sources. The 12-image harbor test counts can be read from the evaluation outputs, although its YAML and membership manifest are absent. The 11-image constructed shipboard dataset was recounted directly from preserved files, but no result matched to the final V0/V1 checkpoints is available.

The compositing rule, alpha-derived annotation, and human acceptance step are defined in Sections 3.5 and 4.2. Hosts, foregrounds, classes, and semantic placement were selected by the operator; the audited interface provides both manual placement and an optional user-

drawn Smart Zone. The software saved the accepted image and synchronized label. The available records provide aggregate counts, but not the exact historical generating revision, asset-to-composite mapping, per-instance placement mode, or transformation settings. The records cannot confirm Smart Zone use for every V1 sample, and the data do not support component-level causal attribution.

A.4 Supplementary Experiment Results

Table 27 summarizes the formal detector comparisons and excludes abandoned, debugging, exploratory, and unexecuted runs. Chapter 5 provides the main interpretation.

Table 27: Formal detector experiment registry.

Experiment family	Conditions and models	Fixed evaluation reference	Main-text location
Architecture comparison	Condition A; YOLO11n, three standalone variants, and four combinations	Corrected and frozen harbor-four evaluation	Section 5.2.2
Annotation correction	A/B; YOLO11n, EfficientViM Block, ASFF, and DySample	Same corrected validation and test labels in each pair	Section 5.3.3
Combined targeted supervision	V0/V1; YOLO11n, ASFF, and DySample	Same final standard evaluation outputs; retained harbor record; constructed-set metrics pending checkpoint reconciliation	Section 5.4.3

A.4.1 Historical Airplane-Context Evaluation Record

This section documents the airplane-only C0/C1 experiment that was formerly presented in the main text. It is not part of the final V0/V1 comparison. The historical C0 checkpoint is also referenced by the alias B0, and C1 by D0; these aliases do not define additional scientific conditions.

The historical dataset builder verifies that C1 differs from C0 only through 34 replaced training hosts and 61 added airplane boxes; it contains no added large vehicle or small vehicle labels. It cannot be the final V1 dataset, which contains all three targeted-supervision components. The historical standard test also contains 742 images and 17,990 instances, including 1,466 airplanes, whereas the final V0/V1 standard output contains 1,087 images and 11,374 instances, including 938 airplanes.

The historical evaluation used the test split at image size 960, batch size 16, confidence

Table 28: Registered identities of the historical airplane-context checkpoints.

Field	Historical C0 (alias B0)	Historical C1 (alias D0)
Experiment ID	`C0_joint7_corrected_no_cp_yolo11n`	`C1_joint7_corrected_train_cp_yolo11n`
Dataset snapshot	`joint_v2_corrected_no_cp`	`joint_v2_corrected_train_cp`
Model and initialization	YOLO11n from `yolo11n.pt`	YOLO11n from `yolo11n.pt`
Training / best epoch	150 / 138	150 / 130
Seed	0	0
Best checkpoint SHA-256 prefix	`a2dc20fc7dffbf56`	`b4eb8f3c64b2a1cf`
Training distinction	No shipboard-aircraft compositing	34 replaced hosts and 61 added airplane boxes only

0.001, NMS IoU 0.7, and at most 300 detections per image. The same two checkpoints were evaluated on the 742-image standard test and the 11-image constructed shipboard test. Table 29 reproduces these historical values for provenance only.

Table 29: Historical airplane-context results, separate from the final V0/V1 experiment.

Evaluation	Historical condition	Instances	Recall	AP@50	AP@50–95
Standard airplane	C0/B0	1,466	0.25363	0.46220	0.23123
Standard airplane	C1/D0	1,466	0.29048	0.42915	0.20069
Constructed shipboard airplane	C0/B0	25	0.00000	0.00000	0.00000
Constructed shipboard airplane	C1/D0	25	0.16000	0.22690	0.12807

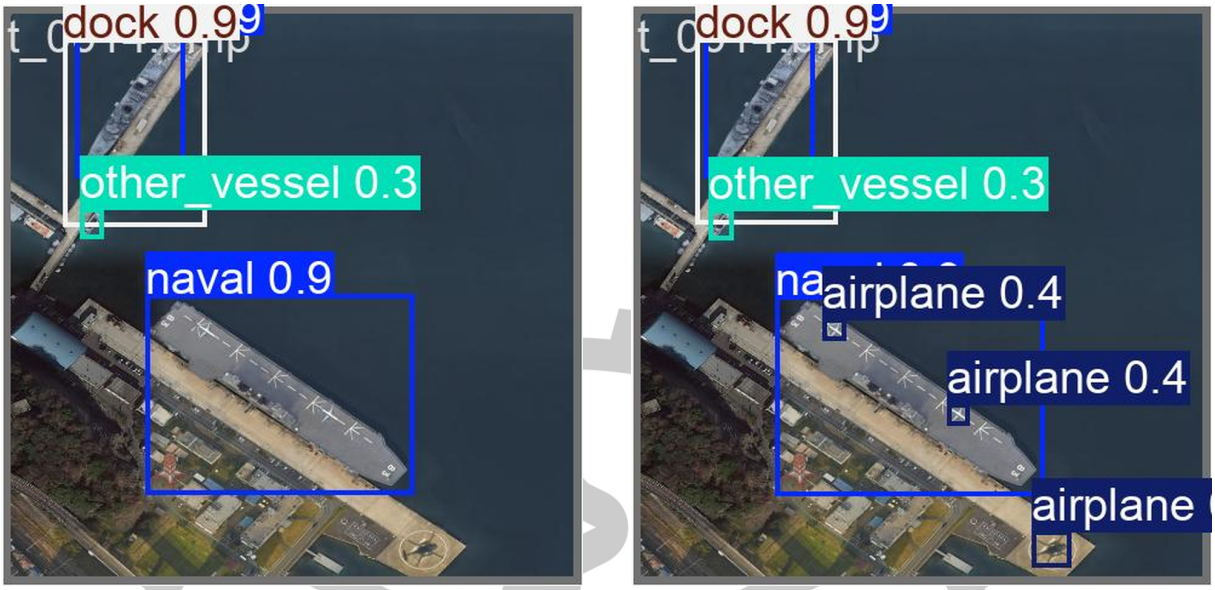
These values are not directly comparable with the final standard-test rows in Chapter 5 because both the training intervention and evaluation population differ. The external experiment archive stores the resolved run arguments, epoch-level logs, selected checkpoints, evaluation summaries, predictions, curves, confusion matrices, and verification manifests.

Figure 15 shows one qualitative example from the historical checkpoints; it is not a V0/V1 result.

A.4.2 Historical Pre-Audit Results

The Old GT results document the architecture comparison that preceded the corrected evaluation reference. They are not a competing leaderboard for Condition A and are not used to estimate the effect of annotation correction.

The separate three-seed Old GT baseline provides historical context for single-run variabil-



(a) Historical C0/B0 prediction.

(b) Historical C1/D0 prediction.

Figure 15: Historical airplane-context predictions on one constructed host, shown for provenance and separate from the final V0/V1 experiment.

Table 30: Historical pre-audit Old GT results for the harbor-four task.

Model	Precision	Recall	mAP@50	mAP@50–95
YOLO11n	0.802	0.726	0.763	0.616
+ EfficientViM Block	0.785	0.729	0.766	0.619
+ DySample	0.778	0.737	0.772	0.626
+ ASFF	0.804	0.717	0.758	0.619
+ EfficientViM Block + DySample	0.776	0.725	0.755	0.606
+ EfficientViM Block + ASFF	0.800	0.719	0.757	0.612
+ DySample + ASFF	0.803	0.721	0.757	0.615
+ EfficientViM Block + DySample + ASFF	0.784	0.721	0.757	0.613

ity. It does not estimate variability for Condition A, Condition B, V0/V1, or C0/C1.

Table 31: Historical Old GT YOLO11n variability under three random seeds.

Baseline run	Precision	Recall	mAP@50	mAP@50–95
`seed=0`	0.8020	0.7260	0.7630	0.6160
`seed=1`	0.7926	0.7262	0.7597	0.6189
`seed=2`	0.7996	0.7267	0.7592	0.6116
Mean	0.7981	0.7263	0.7606	0.6155
Standard deviation	0.0049	0.0003	0.0021	0.0036
Range	0.0094	0.0007	0.0038	0.0072

A.4.3 Supplementary Annotation-Correction Confusion Matrices

Figures 16, 17, and 18 provide the EfficientViM Block, ASFF, and DySample Condition A/B matrices. All pairs use the same corrected evaluation reference; each condition contains 1,464 naval, 1,624 merchant, 198 dock, and 389 ground-truth instances for other vessels, for 3,675 objects in total. Columns are true classes, rows are predicted classes, and the displayed entries are raw counts.

Table 32: Aggregate counts from the Condition A/B confusion matrices. Background FP and FN are reported separately from cross-class confusion.

Model	Condition	TP	Background FP	Background FN	Cross-class confusion
YOLO11n	A	2,924	576	483	268
YOLO11n	B	2,980	675	435	260
EfficientViM Block	A	2,923	602	485	267
EfficientViM Block	B	2,940	651	468	267
ASFF	A	2,909	609	484	282
ASFF	B	2,909	602	460	306
DySample	A	2,896	683	492	287
DySample	B	2,916	639	491	268

A.4.4 Supplementary Harbor Vehicle Confusion Matrices

Figures 19 and 20 provide the complete ASFF and DySample matrices for the 12-image harbor-specific V0/V1 test. The corresponding YOLO11n pair and interpretation are provided in Section 5.4.4. These matrices use the harbor-specific evaluation and must not be read as global seven-class matrices or as C0/C1 aircraft results.

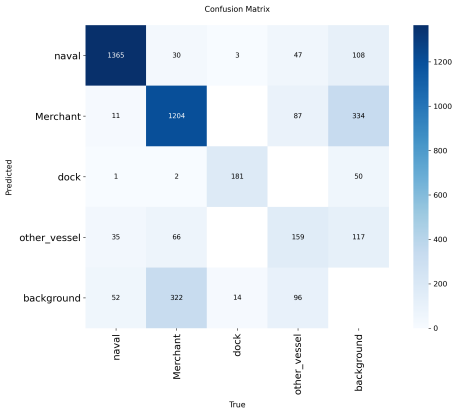


(a) EfficientViM Block under Condition A.



(b) EfficientViM Block under Condition B.

Figure 16: EfficientViM Block confusion matrices under Conditions A and B using the fixed corrected evaluation reference.

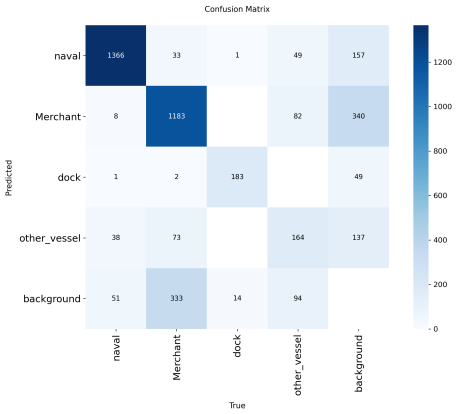


(a) ASFF under Condition A.



(b) ASFF under Condition B.

Figure 17: ASFF confusion matrices under Conditions A and B using the fixed corrected evaluation reference.

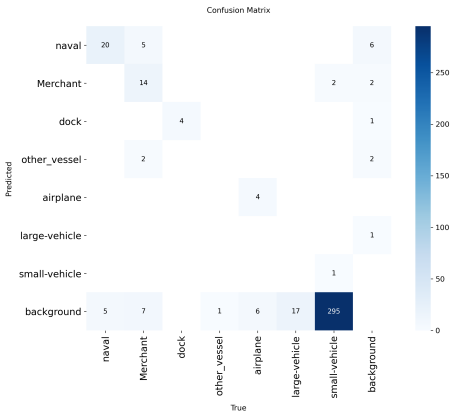


(a) DySample under Condition A.

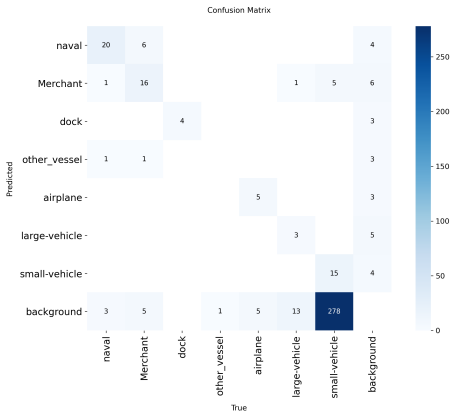


(b) DySample under Condition B.

Figure 18: DySample confusion matrices under Conditions A and B using the fixed corrected evaluation reference.

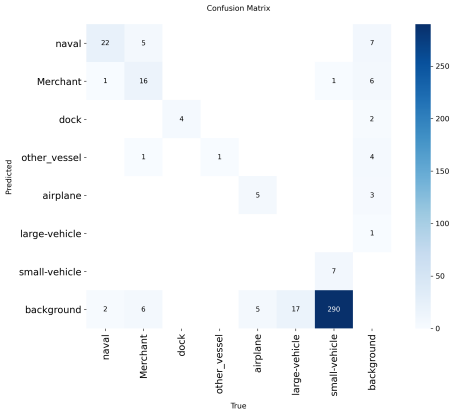


(a) ASFF under V0.

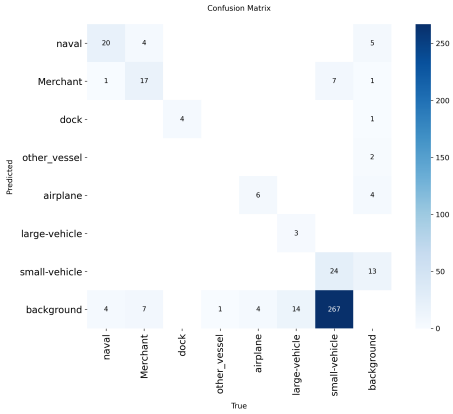


(b) ASFF under V1.

Figure 19: ASFF confusion matrices under V0 and V1 on the 12-image harbor-specific test.



(a) DySample under V0.



(b) DySample under V1.

Figure 20: DySample confusion matrices under V0 and V1 on the 12-image harbor-specific test.

A.5 Research-Support System Reproducibility

Table 33 summarizes the implementation scope of the two research-support systems. These functions support the stated workflows but do not establish detector correctness, operational readiness, or autonomous semantic judgement.

Table 33: Compact implementation record for the two research-support systems.

Record	Verified or source-supported functions	Interpretation limit
SYS1	Dataset import and inspection, annotation review, RGBA foreground handling, manual and Smart Zone placement, synchronized labels, and export in the Dataset Construction and Analysis System.	Source- and interface-supported; the exact historical generating revision was not frozen.
SYS2	Registered-model HBB still-image inference, provenance-bearing results, visualization and history, auxiliary local analysis, and Markdown/PDF export in the web platform.	Verified local workflow; no production, security, throughput, detector-accuracy, or report-quality claim.

SYS1 is associated with the `https://github.com/samhung1205/DetectionData_Engine` source repository. Its operator-defined placement and synchronized annotation behavior are described in Chapters 3 and 4. The source demonstrates the implemented workflow, while host suitability, foreground suitability, transform selection, and final acceptance remain human decisions.

SYS2 is tied to the recorded source revision `3c7deee`. The server-side registry resolves a stable model key to an approved checkpoint and class map, checks the checkpoint and any required custom-module source fingerprints, and runs only the registered horizontal-bounding-box models. Each stored result associates the input, model identity, class-map snapshot, confidence and IoU thresholds, image dimensions, predicted classes, confidences, boxes, and inference time. This association permits later inspection without relying on a local checkpoint path supplied by the browser.

The stored result supplies visualization, history, and reports. Auxiliary generated text reads selected fields from the result, cannot modify detector outputs or ground truth, and requires human review. Exact dependency captures, maintenance records, endpoint lists, internal model keys, and complete fingerprints remain in the external system archive.

A.6 Vessel-Type Assistant Supplementary Evaluation

VC1 calibration and one-time VC2 confirmation assess the frozen supplied-box vessel-type assistant. Earlier development-set diagnostics were used only during model development and are not included in the confirmation results. All frozen evaluation files were checked against the artifact manifest.

The assistant outputs a ranking over `naval`, `merchant`, and `other_vessel`; `dock` is excluded because it is not a vessel type. Because it receives a researcher-supplied box, it does not evaluate object localization. Table 34 summarizes the configuration held fixed for VC1 and VC2.

Table 34: Frozen configuration of the vessel-type annotation assistant.

Item	Frozen configuration
Output classes	`naval`, `merchant`, and `other_vessel`
Input views	Tight supplied-box crop and 25% context crop, each resized to 224×224
Representation	Frozen ImageNet ResNet-50 features [30, 31] and five independently trained multi-view heads
Decision layer	Frozen five-head ensemble and class-bias policy followed by the frozen naval-rescue rule
Training control	AdamW, learning rate 0.00055, weight decay 0.0003, cosine schedule, at most 70 epochs, batch size 256, and patience 12

The fresh evaluation sources comprise FAIR1M-2.0 validation images and a selected naval supplement from FAIR1M train [32]. Exact, perceptual, and multiscale geometric screening found no unapproved overlap between the 319 selected scene groups and the 3,435 registered development images. Independent reference decisions covered 2,000 objects; adjudication produced 1,907 trusted vessels and 93 exclusions. The calibration and confirmation scene groups are disjoint, and neither subset was used to revise the feature encoder, heads, bias policy, or rescue rule.

Table 35: Roles of the frozen vessel-assistant evaluation subsets.

Subset	Population	Analysis role
VC1 calibration	336 objects; 49 scene groups	Fit one scalar temperature; no model, policy, label, or threshold revision.
VC2 confirmation	1,571 objects; 269 scene groups	One evaluation of the unchanged system; no refitting or iterative model selection.

VC1 fitted one positive scalar temperature by minimizing negative log-likelihood [33]. The

frozen value is $T = 2.861514$; it changes no top-1 decision, and no automatic acceptance or abstention threshold is used. Table 36 gives the supplementary calibration results.

Table 36: Calibration metrics before and after the frozen temperature transform.

Subset	ECE before	ECE after	NLL before	NLL after	Brier before	Brier after
VC1	0.1957	0.0574	1.2039	1.0238	0.6813	0.6142
VC2	0.1574	0.0622	1.1671	1.0264	0.6737	0.6190

The predefined confirmation analyses are summarized in Table 37. Main-text accuracy values and their interpretation are reported in Section 5.3.2.

Table 37: Predefined VC2 analysis tracks and interpretation limits.

Track	Population	Evaluation role
A	1,571 trusted vessels	Compares full-image YOLO with the supplied-box assistant while explicitly accounting for the input asymmetry.
B	807 one-to-one matched YOLO objects	Records paired outcomes with exact McNemar tests and scene-group bootstrap intervals.
C	764 YOLO misses	Evaluates assistant top-1 and top-2 ranking after the reference box is supplied; it is not a localization rescue test.

Frozen evaluation artifacts and fingerprints remain in the external archive. Every assistant ranking is a proposal that requires human confirmation before it enters an annotation record.